

SURREAL AND SURCOMPLEX RESEARCH

Holonomic Rigidity for Entire Hahn Functions

A sharp order-unit criterion for dilations,
with uniform exclusion bounds

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Abstract

Fix a nonzero set-sized ordered abelian group Γ and the complex Hahn field $K_\Gamma = \mathbb{C}((t^\Gamma))$. *Entire* means strong Hahn summability of the evaluation family at every point of this one field. We prove that every strongly entire D-finite power series is a polynomial, in one or in finitely many variables, and that a fixed differential operator already excludes every nonterminating formal solution on an explicit exterior valuation region, with a common degree bound.

For dilations there is an exact alternative. If $q \in K_\Gamma^\times$ has infinite multiplicative order, a nonpolynomial strongly entire solution of a nonzero linear q -difference equation with polynomial coefficients exists if and only if $v(q) \neq 0$ and $|v(q)|$ is an order unit of Γ . A nonzero valuation is not enough: its integer multiples must be cofinal among all scales. The proof covers units whose residues are roots of unity, and coefficients and parameters with arbitrary well-ordered Hahn supports. A partial theta series proves sharpness, and its exact strong evaluation domain is computed for an arbitrary Hahn parameter. Quantitatively, the recurrence supplies an exclusion bound that is attained on its own boundary, and a coarsened exclusion bound valid for every noncofinal dilation. A mixed differential–dilation extension is proved for nonzero noncofinal dilation valuations.

The common mechanism is an elementary finite-step escape chain: bounded valuation costs in a coefficient recurrence force a subsequence of evaluated leading exponents that never increases. This contradicts strong summability without replacing an arbitrary-rank valuation by a real-valued norm. In an explicit surreal workspace with countable cofinality and no order unit, the series $\sum_{n \geq 0} \omega^{-\omega^n} z^n$ is strongly entire yet satisfies no nonzero linear differential equation and no nonzero linear dilation equation for any nontorsion parameter of that workspace.

Γ is *not* assumed divisible. Divisibility is hypothesised only where it is used, in the three threshold constructions listed in Section 1.4, and each of those statements prints the hypothesis itself.

Status. This is a merged AI-assisted research report with written mathematical proofs and two independent finite exact-arithmetic check suites. The arbitrary-rank classification is offered as a proposed original contribution, not as certified bibliographic priority and not as the resolution of a named published conjecture. The positive-support word lemma has Lean coverage; the main theorem package remains unformalized and has not been independently refereed. The variable derivative used throughout is not the Berarducci–Mantova derivation on surreal coefficients.

Contents

1	The question and the principal results	2
2	Hahn support and what evaluation means	6
3	The finite-step escape mechanism	9
4	Valuations on integer and geometric orbits	11
5	Differential rigidity	14
6	Generic lines and several-variable rigidity	17
7	Dilation equations and the necessity of a cofinal scale	18
8	Partial theta series: sharp witness and exact domain	21
9	Torsion parameters and why they are excluded	23
10	An extension to mixed differential–dilation equations	24
11	Surreal workspaces that separate the theories	25
12	Coefficient fields, degenerate value groups, characteristic	28
13	Quantitative certificates and symbolic representation	28
14	The two finite check suites	30
15	Originality assessment and questions left open	32
16	Conclusion	34
A	Notation	35
B	Proof-dependency and hypothesis audit	35
C	Reproducing the document and the finite checks	36
	References	37

1 The question and the principal results

1.1 Why the question is scale sensitive

A familiar way to specify a special function is to give a finite linear differential equation. Another is to give a finite relation between its values at $z, qz, q^2z, \dots$. Over the complex numbers both descriptions produce nonpolynomial entire functions. Over a surreal Hahn workspace, “entire” asks for evaluation not only at ordinary complex constants but at every infinite and infinitesimal scale available in that workspace. In a Hahn field an infinite expression is legal only when its supports satisfy an order condition and a finiteness condition, and an equation can force either condition to fail at a sufficiently large argument. The question of this report is what a finite linear equation can accommodate under that stronger global requirement.

The answer depends decisively on which operator is used, and for dilation equations it depends further on whether the valuation of the particular parameter reaches every scale of the value group.

1.2 Conventions

We work with a fixed ordered abelian group Γ and

$$K_\Gamma = \mathbb{C}((t^\Gamma)).$$

An element of K_Γ is a formal sum $a = \sum_\gamma a_\gamma t^\gamma$ with $a_\gamma \in \mathbb{C}$ whose support $\text{supp}(a) = \{\gamma : a_\gamma \neq 0\}$ is a well-ordered set. For $a \neq 0$ put

$$v(a) = \min \text{supp}(a), \quad \text{lc}(a) = a_{v(a)} \in \mathbb{C}^\times, \quad \text{res}(a) = \text{lc}(a) \text{ when } v(a) = 0,$$

and put $v(0) = +\infty$ only where convenient; this symbol is never subtracted. Products and sums are the usual Hahn operations, with

$$v(ab) = v(a) + v(b), \quad v(a + b) \geq \min\{v(a), v(b)\}, \tag{1.1}$$

with equality in the second law when the two valuations differ. The coefficient field $\mathbb{C}$ is trivially valued, so every nonzero ordinary integer and every factorial has valuation zero. Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$, and $|\lambda|$ means $\max(\lambda, -\lambda)$ in the ordered group.

Convention 1.1 (Standing hypothesis on the value group). Γ is a nonzero set-sized ordered abelian group. Divisibility is *not* part of this standing hypothesis. Every statement below that needs divisibility says so in its own statement; those statements are collected in Section 1.4.

Definition 1.2 (Strong summability, evaluation domain, entireness). A set-indexed family $(b_i)_{i \in I}$ in K_Γ is *strongly Hahn summable* if $\bigcup_i \text{supp}(b_i)$ is well ordered and each $\gamma \in \Gamma$ lies in the supports of only finitely many members. Its sum is then defined coefficientwise, by a finite sum in $\mathbb{C}$ at each exponent. For $f = \sum_{n \geq 0} a_n z^n \in K_\Gamma[[z]]$ let $\text{Dom}_{\text{str}}(f)$ be the set of $x \in K_\Gamma$ for which $(a_n x^n)_{n \geq 0}$ is strongly summable, and let

$$\mathcal{E}_\Gamma = \{f : \text{Dom}_{\text{str}}(f) = K_\Gamma\}.$$

In d variables the family to be tested is $(a_\alpha x^\alpha)_{\alpha \in \mathbb{N}^d}$.

The word *entire* always means Definition 1.2 relative to one fixed workspace. It does not mean that a complex coefficient function is entire, it does not mean convergence in the fine topology

inherited from the full surreal class, and it is never a claim about all of $\mathbf{No}[i]$. The definition also concerns the family *before* cancellation between different indices: one cannot first cancel a nonsummable family and then declare it summable, and convergence of a numerical coefficient series does not license infinitely many contributions at one Hahn exponent.

Definition 1.3 (Order unit). A positive $\mu \in \Gamma$ is an *order unit* if for every $\gamma \in \Gamma$ there is $N \in \mathbb{N}$ with $|\gamma| \leq N\mu$; equivalently, if the positive integer multiples of μ are cofinal in Γ . Zero is not an order unit.

Definition 1.4 (The two operator classes). Let D_z be the formal derivative in z , so $D_z(a) = 0$ for every $a \in K_\Gamma$, and for $q \in K_\Gamma^\times$ let $\sigma_q f(z) = f(qz)$. A series $f \in K_\Gamma[[z]]$ is *D-finite over $K_\Gamma(z)$* if

$$\sum_{r=0}^R p_r(z) D_z^r f(z) = 0, \quad p_r \in K_\Gamma[z], \quad (p_0, \dots, p_R) \neq 0, \quad (1.2)$$

and *linearly q -difference finite* if

$$\sum_{\ell=0}^r p_\ell(z) f(q^\ell z) = 0, \quad p_\ell \in K_\Gamma[z], \quad (p_0, \dots, p_r) \neq 0. \quad (1.3)$$

These are formal identities in $K_\Gamma[[z]]$, not assertions about limits of partial sums. Rational coefficients give the same notions after denominators are cleared.

We say “linearly q -difference finite” rather than silently identifying several meanings of “ q -holonomic”. In the sequence literature q is often an indeterminate with coefficients in $\mathbb{Q}(q)$; here q is a *fixed* Hahn element, possibly with infinitely many terms. The differential terminology is the usual characteristic-zero one [1]. *Nontorsion* and “of infinite multiplicative order” are used interchangeably.

1.3 The principal results

Theorem A (Differential rigidity, with a uniform exterior obstruction). *Let Γ be as in Convention 1.1 and let $0 \neq L = \sum_{r,k} c_{r,k} z^k D_z^r$ be a nonzero linear differential operator with coefficients in $K_\Gamma[z]$. There are $B_L \in \Gamma$ with $B_L \geq 0$ and $N_L \in \mathbb{N}$, both determined by the finitely many coefficients of L , such that every formal solution of $Lf = 0$ that is strongly evaluable at even one $x \neq 0$ with $v(x) \leq -B_L$ has degree less than N_L . Consequently*

$$\{f \in \mathcal{E}_\Gamma : f \text{ is } D\text{-finite over } K_\Gamma(z)\} = K_\Gamma[z].$$

Theorem B (Sharp dilation criterion). *Let Γ be as in Convention 1.1 and let $q \in K_\Gamma^\times$ have infinite multiplicative order. The following are equivalent.*

- (i) *There is a nonpolynomial $f \in \mathcal{E}_\Gamma$ that is linearly q -difference finite over $K_\Gamma(z)$.*
- (ii) *$v(q) \neq 0$ and $|v(q)|$ is an order unit of Γ .*

When (ii) fails, every fixed equation (1.3) has a uniform exterior obstruction and a uniform polynomial degree bound, exactly as in Theorem A. When (ii) holds, an explicit partial theta series with parameter q or q^{-1} is a nonpolynomial entire solution.

Theorem C (Several-variable rigidity). *Let Γ be as in Convention 1.1 and let $F \in K_\Gamma[[z_1, \dots, z_d]]$ be strongly Hahn summable at every point of K_Γ^d . If the mixed partial derivatives of F span a finite-dimensional vector space over $K_\Gamma(z_1, \dots, z_d)$, then F is a polynomial.*

The hypotheses separate two obstructions. Differential equations have bounded valuation costs after passing to their coefficient recurrence. Dilation equations can have costs increasing linearly with the index, but only along the single scale $v(q)$; a scale that is not an order unit cannot reach all of Γ , even if its multiples are unbounded within their own Archimedean component. Strong entireness detects that failure.

Beyond these three, the report proves: a mixed differential–dilation rigidity theorem at every nonzero noncofinal dilation scale (Theorem 10.2); a recurrence exclusion bound that is *attained*, boundary included, by an explicit formal exponential (Corollary 3.5 and Proposition 5.3); a coarsened exclusion bound valid at every nonzero noncofinal dilation valuation (Theorem 7.4); and the exact strong evaluation domain of the partial theta series for an arbitrary Hahn parameter, including arbitrary tails (Theorem 8.1).

1.4 Where divisibility of Γ is used

The core arguments — the escape mechanism, the orbit valuation lemmas, the two conversions, the theta domain theorem, and all of Theorems A–C — use only Convention 1.1. Divisibility of Γ enters in exactly three threshold constructions, with the hypothesis stated where it is used.

- (1) The refined recurrence threshold B_{rec} of (3.4), whose defining maximum divides a valuation difference by the jump length j . Corollary 3.5 states it under divisibility; Corollary 3.4 is the divisibility-free predecessor, with a coarser threshold and the same qualitative conclusion.
- (2) The refined coarsened threshold $\overline{B}_{\text{rec}}$ of (7.4), for the same reason, in the quotient group. Theorem 7.4 states the divisibility-free form and the divisible refinement separately.
- (3) The worked sparse threshold in Example 13.1, an instance of (1).

The torsion eigenspace equivalence in Proposition 9.1 needs no divisibility: inward stability of strong evaluation replaces the root-extraction argument of the source manuscripts.

No theorem in this report states a conclusion under Convention 1.1 while resting on a proof step that divides by an integer in Γ . The division-free predecessors in (1) and (2) are weaker in the constant only, not in the conclusion.

1.5 Position relative to the collection and the literature

The repository snapshot consulted by both source manuscripts is

a3124af79f66b8b9c196d76b4cbc5ac3938907c4.

The companion entire-function report. The collection already contains a merged report on cofinality, factorization and scalar extension for entire Hahn functions at arbitrary rank [12]. It fixes the same class $\mathcal{E}_\Gamma$ over the same kind of workspace, classifies the ring structure of $\mathcal{E}_\Gamma$, and computes the exact strong evaluation domain after an enlargement of the value group. None of its ring-theoretic conclusions is used to prove anything here: the present proofs start from Hahn support arithmetic. Two points of contact matter.

First, we do not reprove the order-unit coarsening machinery. That report’s Lemma 9.1, labelled `ent:lem:coarsening`, states that if $h > 0$ is an order unit of Γ there is an order-preserving homomorphism $\pi : \Gamma \rightarrow \mathbb{R}$ with $\pi(h) = 1$ whose kernel is the convex subgroup $H = \{\gamma : n|\gamma| <$

h for all $n \geq 1$ }, so that $|a|_\pi = \exp(-\pi(v(a)))$ is a nontrivial real-valued non-Archimedean absolute value inducing the intrinsic valuation topology. That is exactly the sense in which both source manuscripts of this report use the phrase “coarsening”, and we cite it rather than repeat it; see Remark 2.6.

Second, and this is easy to get wrong: *the two order-unit dichotomies are not the same theorem* — the companion report’s order unit decides Hermite interpolation and the Bézout property of the ring $\mathcal{E}_\Gamma$ (its Theorem 9.3), whereas the order unit here decides which annihilating operators a nonpolynomial member of $\mathcal{E}_\Gamma$ can satisfy. Neither dichotomy implies the other, and the coincidence of the invariant is not a coincidence of statement. In particular the existence of nonpolynomial entire functions at countable cofinality, and the fact that $\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q}\omega^j$ has countable cofinality and no order unit, are background from that report [12], not targets of this one.

The collection’s differential-equations report uses a derivation on surreal *scalars* [13]; here D_z kills every scalar and differentiates only the formal variable, and no theorem about that scalar derivation is restated as ordinary D-finite rigidity. The rank-one analytic report [14] already contains the quadratic-exponent example $\sum t^{n^2} z^n$; our partial theta witness is explicitly classical and is not counted as a new function.

External literature. Stanley’s D-finite/P-recursive correspondence is the standard algebraic starting point [1]. We make the coefficient conversion explicit because the valuations of the resulting recurrence, rather than ordinary coefficient growth, drive the proof. Ramis treats growth of ordinary complex entire solutions of linear q -difference equations and also studies simultaneous differential and difference constraints [2]; the arbitrary-rank strong summability condition here is a different global condition, and none of its order-unit conclusions is inferred from an ordinary complex growth estimate. Garoufalidis proves eventual quadratic quasi-polynomial degree behaviour for q -holonomic sequences in a rational-function and Puiseux setting [3]; that is an important antecedent to the idea that dilation equations constrain coefficient scales, we do not claim to originate it, and we do not strengthen its quadratic quasi-polynomial conclusion. Di Vizio’s ultrametric Maillet–Malgrange theorem treats Gevrey behaviour of formal solutions of nonlinear ultrametric q -difference equations, including the valuation-zero case with Diophantine conditions [4]; we make no nonlinear generalization of it. The partial theta series belongs to the classical partial-theta literature [5], and Higman’s finite-word theorem is a standard external ingredient in the positive-support monoid lemma [6].

Terminology needs care in two further directions. Hahn meromorphic functions in the analytic setting of Müller and Strohmaier involve a convergence framework different from the formal strong evaluation used here [7]. The Berarducci–Mantova derivation differentiates surreal coefficients themselves and is not D_z [8]. Work on operation-closed surreal subfields gives context for set-sized workspaces [9], and the Conway normal-form conventions used in Section 2.4 follow current surreal field work [10]; closure under a scalar derivation is assumed in no theorem below.

Both source inspections were targeted: repository documentation, the report catalogue, the entire-function report, and the primary sources named above. Neither was a line-by-line audit of every repository manuscript or an exhaustive bibliography search, and a repository content search for `holonomic` returning no matches is not proof of absence. Section 15 records exactly what is and is not claimed.

2 Hahn support and what evaluation means

2.1 Support calculus

Lemma 2.1 (Hahn–Neumann support calculus and the positive monoid). *Let A, B be well-ordered subsets of an ordered abelian group.*

- (1) *$A + B$ is well ordered, and each γ has only finitely many representations $\gamma = a + b$ with $(a, b) \in A \times B$.*
- (2) *If S is a well-ordered subset of $\Gamma_{>0}$, the additive monoid*

$$M(S) = \{s_1 + \cdots + s_k : k \geq 0, s_i \in S\}$$

is well ordered, and each of its elements has only finitely many representations as a finite word in letters from S .

- (3) *Multiplication by a fixed Hahn element preserves strong summability, and multiplication by a nonzero one preserves and reflects it.*

Proof. An infinite sequence in a well-ordered set has an infinite nondecreasing subsequence. Given a strictly decreasing sequence $a_n + b_n$, pass first to a subsequence with a_n nondecreasing and then to one with b_n nondecreasing, a contradiction. If infinitely many distinct pairs sum to a fixed γ , the distinct first coordinates contain an increasing sequence, forcing a decreasing sequence of second coordinates. This is part (1).

For part (2), order finite words over S by subsequence embedding with coordinatewise increase. Higman’s theorem says that words over a well-quasi-ordered alphabet are well quasi ordered [6]. An embedding cannot decrease the sum, and increases it strictly unless the two words are identical, since inserted letters are positive and a strict coordinate increase is positive. A descending sequence of sums therefore gives a bad sequence of words, and so do infinitely many distinct words with one fixed sum. Both are impossible.

For part (3), let A be the union of the supports of a strongly summable family and B the support of the multiplier. The new supports lie in $A + B$. Part (1) leaves finitely many admissible pairs at a specified exponent, and for each first coordinate only finitely many family members contribute. For a nonzero multiplier apply the same to its inverse. $\square$

In part (2) words carry their order; a fixed finite multiset has only finitely many permutations, so the commutative monoid version follows. The argument does not assume that S is bounded below by an order unit, and remains valid at arbitrary valuation rank.

Lemma 2.2 (The nonincreasing leading-exponent obstruction). *A family of nonzero Hahn elements possessing an infinite subsequence whose leading exponents are nonincreasing is not strongly summable.*

Proof. If that sequence of leading exponents takes infinitely many distinct values it has an infinite strictly decreasing subsequence, so the union of the supports is not well ordered. Otherwise it is eventually constant, and one exponent lies in the supports of infinitely many members, so local finiteness fails. $\square$

The second alternative is essential and is what makes the exclusion boundary below a *closed* region. Merely excluding a descending support does not establish strong summability: the family $(1 \cdot z^n)_{n \geq 0}$ at $z = 1$ has support union $\{0\}$, yet every summand contributes at exponent zero.

Lemma 2.3 (Two sufficient certificates). *Let $(b_n)_{n \geq 0}$ be a family in K_Γ .*

- (a) *If $v(b_n)$ tends cofinally to $+\infty$, meaning that for each $\beta \in \Gamma$ all but finitely many nonzero b_n satisfy $v(b_n) > \beta$, then (b_n) is strongly summable.*
- (b) *If $\text{supp}(b_n) \subseteq h_n + M(S)$ for a well-ordered $S \subseteq \Gamma_{>0}$, where the set of values of h_n is well ordered and each value is taken by only finitely many indices, then (b_n) is strongly summable.*

Moreover, grouping a strongly summable family into finite blocks preserves strong summability and its sum.

Proof. For (a) fix β . Only finitely many supports meet $(-\infty, \beta]$, and a finite union of well-ordered subsets of a linear order is well ordered. A descending sequence in the full union would lie below its first member, hence inside such a finite union. The same finiteness bounds the multiplicity of an exponent. For (b) use parts (1) and (2) of Lemma 2.1: a fixed target exponent admits finitely many pairs $(h, m) \in \{h_n\} \times M(S)$ with $h + m$ equal to it, and finitely many indices carry each such h ; actual supports can only be smaller than the indicated sets. For the last assertion, each block support is contained in the union of the old supports and the blocks contributing at any exponent form a finite set, while coefficientwise associativity is finite. $\square$

Criterion (a) and criterion (b) are genuinely different tools and both are used below. Criterion (a) is the shorter route whenever the term valuations really do escape every bound. Criterion (b) is what survives at the boundary, where the term valuations increase strictly but remain trapped below a fixed element of Γ ; Theorem 8.1 is exactly such a case, and no leading-valuation argument alone can decide it.

Lemma 2.4 (Inward stability of strong evaluation). *Let $f = \sum_n a_n z^n \in K_\Gamma[[z]]$, and let $x, y \in K_\Gamma^\times$ with $v(y) \geq v(x)$. If $x \in \text{Dom}_{\text{str}}(f)$, then $y \in \text{Dom}_{\text{str}}(f)$. Thus membership of a nonzero argument in $\text{Dom}_{\text{str}}(f)$ depends only on its valuation, and entireness can be tested on monomial arguments.*

Proof. Put $b_n = a_n x^n$ and $u = y/x$, so (b_n) is strongly summable and $v(u) = \eta \geq 0$. Write $u = c t^\eta(1 + w)$, where $c \in \mathbb{C}^\times$ and w is zero or has positive valuation. Let S be $\text{supp}(w)$, with η adjoined when $\eta > 0$, and put $M = M(S)$. This is a well-ordered positive alphabet, so Lemma 2.1 makes M well ordered. Every finite power u^n has support in M .

Let $A = \bigcup_n \text{supp}(b_n)$. The supports of $a_n y^n = b_n u^n$ lie in $A + M$, a well-ordered set. For a fixed exponent γ there are only finitely many pairs $(\alpha, \mu) \in A \times M$ with $\alpha + \mu = \gamma$. For each such α , strong summability of (b_n) leaves only finitely many indices n . Hence γ occurs in only finitely many supports of $b_n u^n$, proving strong summability. Equal valuations allow the argument in both directions; compare an arbitrary nonzero y with $t^{v(y)}$ to obtain the monomial test. Evaluation at zero is always finite. $\square$

The statement fixes the full coefficient series a_n . It does not say that their leading valuations alone determine the domain: higher coefficient tails can still matter at a boundary scale.

2.2 Positive valuation is not cofinal decay

For $\beta > 0$ write

$$\Delta_\beta = \{\gamma \in \Gamma : |\gamma| \leq N\beta \text{ for some } N \in \mathbb{N}\}, \quad (2.1)$$

the smallest convex subgroup containing β . It is all of Γ exactly when β is an order unit. If it is proper, there is $\gamma > 0$ outside it, and then

$$\gamma > N\beta \quad \text{for every ordinary } N \in \mathbb{N}, \quad (2.2)$$

since a positive element bounded by some $N\beta$ would lie in Δ_β .

Remark 2.5 (Sufficiency is not necessity). Let $0 < \epsilon < H$ in Γ with $n\epsilon < H$ for every ordinary integer n ; for instance $\Gamma = \mathbb{Q}H \oplus \mathbb{Q}\epsilon$ ordered by the highest nonzero coordinate. The family $(t^{n\epsilon})_{n \geq 0}$ is strongly summable, since its support is an increasing sequence and each exponent occurs once, and its sum is the Hahn element $(1 - t^\epsilon)^{-1}$. Yet its valuations do not tend cofinally to $+\infty$. This is the most elementary warning against identifying positive valuation with topological nilpotence, and it is why criterion (b) of Lemma 2.3 is needed at all. No appeal to convergence of partial sums at the H scale is involved.

Remark 2.6 (The real-valued coarsening is imported, not reproved). When Γ does have an order unit h , the companion report [12, Lemma 9.1, `ent:lem:coarsening`] supplies an order-preserving $\pi : \Gamma \rightarrow \mathbb{R}$ with $\pi(h) = 1$ and kernel the convex subgroup $H = \{\gamma : n|\gamma| < h \text{ for all } n \geq 1\}$, hence a nontrivial real-valued non-Archimedean absolute value $|a|_\pi = \exp(-\pi(v(a)))$ inducing the intrinsic topology. We cite that lemma and do not reprove it. No proof in this report uses it: every argument below is carried out in Γ itself, which is what makes the arguments valid when no order unit exists. The coarsening is mentioned twice, as an optional rank-one picture of the order-unit case (Remark 8.3) and for the warning of Section 1.5 that the interpolation/Bézout dichotomy of that report and the annihilating-operator dichotomy of this one share an invariant but are different theorems.

2.3 The entire class and its closure properties

Polynomials are entire, because all the evaluation families are finite. An ordinary complex entire Taylor series need not be entire in the sense of Definition 1.2: at a monomial argument $x = t^\gamma$ the family

$$\left(\frac{x^n}{n!}\right)_{n \geq 0} \quad (2.3)$$

is strongly summable exactly when $\gamma > 0$, since $\gamma = 0$ puts infinitely many contributions at exponent zero and $\gamma < 0$ gives a descending sequence. Proposition 5.3 computes the exact domain of the corresponding family with an arbitrary Hahn argument.

Proposition 2.7 (Closure). $\mathcal{E}_\Gamma$ is a K_Γ -algebra, $\sigma_q(\mathcal{E}_\Gamma) \subseteq \mathcal{E}_\Gamma$ for every $q \in K_\Gamma^\times$, and $D_z(\mathcal{E}_\Gamma) \subseteq \mathcal{E}_\Gamma$.

Proof. For addition, the evaluation supports lie in the union of two well-ordered sets with finite exponent multiplicities. For multiplication, the doubly indexed product family is strongly summable by Lemma 2.1 (1) and (3), and grouping by ordinary total degree preserves strong summability by the last clause of Lemma 2.3. Dilation is evaluation at qx . For $x \neq 0$ the derivative family is

$$(n+1)a_{n+1}x^n = x^{-1} \cdot (n+1)a_{n+1}x^{n+1}.$$

Discarding one term and multiplying each term by a nonzero ordinary integer enlarges neither supports nor multiplicities, and multiplication by x^{-1} preserves summability by Lemma 2.1 (3). At $x = 0$ only the constant term contributes. Iterate. $\square$

Proposition 2.8 (Faithfulness of evaluation). *If $f \in \mathcal{E}_\Gamma$ vanishes at infinitely many distinct ordinary complex arguments, then $f = 0$ as a formal series. Consequently a linear equation with rational-function coefficients that holds pointwise wherever its coefficients are defined is also a formal equation in the sense of Definition 1.4.*

Proof. Strong evaluation at 1 says that the coefficient family (a_n) has a well-ordered union of supports and is locally finite at every exponent. Hence for each γ

$$P_\gamma(T) = \sum_n (a_n)_\gamma T^n \in \mathbb{C}[T]$$

is a polynomial, and its value at an ordinary complex c is the coefficient of t^γ in $f(c)$. If f vanishes at infinitely many such c , every P_γ has infinitely many roots and is zero, so every a_n is zero. For the last assertion, clear denominators; the left-hand side is in $\mathcal{E}_\Gamma$ by Proposition 2.7, and nonzero polynomial denominators exclude at most finitely many ordinary complex arguments. $\square$

The nonexistence proofs may therefore be run on formal coefficients without weakening their reading as functional obstructions, and no interchange of infinite limits is needed anywhere.

The theorems below use only the valuation laws (1.1), the existence of every monomial t^γ , triviality of the valuation on $\mathbb{C}^\times$, and the support facts of this section. They do not require algebraic closedness, spherical completeness, Weierstrass preparation, or an embedding of Γ into $\mathbb{R}$.

2.4 Embedding the workspace into the surcomplex numbers

If Γ is an additive ordered subgroup of $\mathbf{No}$, the Conway normal-form map

$$\sum_{\gamma \in S} c_\gamma t^\gamma \mapsto \sum_{\gamma \in S} c_\gamma \omega^{-\gamma} \tag{2.4}$$

identifies the real-coefficient Hahn field with a subfield of $\mathbf{No}$ and its complexification with a subfield of $\mathbf{No}[i]$; a well-ordered S becomes a reverse-well-ordered set of Conway exponents. This viewpoint and its set-sized support discipline are standard [9, 10]. Here t^γ is a formal monomial and $\omega^{-\gamma}$ a Conway monomial; the notation does not silently identify every formal power with a choice of surreal exponentiation.

The results may therefore be read twice. Abstractly they concern Hahn fields and need no surreal construction. Concretely they restrict the exact linear descriptions available for surcomplex-valued power series on specified surreal scales. Formula (2.4) is the only bridge used, and no global surcomplex exponential is invoked.

3 The finite-step escape mechanism

The following isolates the mechanism used throughout. It needs no formula for the recurrence coefficients, no asymptotic slope for the valuations, and no assumption that cancellations are rare.

Lemma 3.1 (Escape chain). *Suppose a sequence (a_n) in K_Γ satisfies, for every $n \geq N$,*

$$c_0(n)a_n + \sum_{j=1}^s c_j(n)a_{n+j} = 0, \quad c_0(n) \neq 0, \quad (3.1)$$

and suppose $a_{n_0} \neq 0$ for some $n_0 \geq N$. Then $s \geq 1$ and there is an infinite sequence $n_0 < n_1 < n_2 < \dots$ with $j_r := n_{r+1} - n_r \in \{1, \dots, s\}$, with $a_{n_r} \neq 0$ and $c_{j_r}(n_r) \neq 0$ for every r , and with

$$v(a_{n_{r+1}}) - v(a_{n_r}) \leq v(c_0(n_r)) - v(c_{j_r}(n_r)). \quad (3.2)$$

Proof. Let $n \geq N$ with $a_n \neq 0$. The leftmost term of (3.1) is nonzero, so $s \geq 1$ and at least one later term is nonzero. If every nonzero later term had valuation strictly larger than $v(c_0(n)a_n)$, the leading monomial of the leftmost term could not cancel, contradicting (1.1). So some $j \in \{1, \dots, s\}$ has $c_j(n)a_{n+j} \neq 0$ and

$$v(c_j(n)a_{n+j}) \leq v(c_0(n)a_n),$$

which is (3.2). Taking the least eligible j makes the recursion definite. The successor coefficient is nonzero and its index is again at least N , so the construction never stops. $\square$

Remark 3.2 (The direction of the inequality matters). Solving the recurrence for its *last* coefficient typically yields a *lower* bound on a valuation, which obstructs nothing. The comparison above is made from the first nonzero coefficient forward and produces a later coefficient whose valuation is controlled from above. Cancellation among the other terms can raise the valuation of their sum, but cannot make every term on the right have valuation strictly larger than the single nonzero term on the left. Zeros and arbitrarily large cancellations elsewhere therefore do not invalidate the chain; the chain records only that finite inequality, repeatedly.

Theorem 3.3 (Evaluation exclusion criterion). *In the situation of Lemma 3.1, let $x \in K_\Gamma^\times$ and suppose that for every $n \geq N$ and every $j \in \{1, \dots, s\}$ with $c_j(n) \neq 0$,*

$$v(c_0(n)) - v(c_j(n)) + jv(x) \leq 0. \quad (3.3)$$

If $a_{n_0} \neq 0$ for some $n_0 \geq N$, then $(a_n x^n)_{n \geq 0}$ is not strongly summable. Equivalently: if $(a_n x^n)_{n \geq 0}$ is strongly summable, then $a_n = 0$ for every $n \geq N$.

Proof. Build the chain of Lemma 3.1 from n_0 and add $n_{r+1}v(x)$ to each side of (3.2), using $n_{r+1} = n_r + j_r$:

$$v(a_{n_{r+1}}x^{n_{r+1}}) \leq v(a_{n_r}x^{n_r}) + v(c_0(n_r)) - v(c_{j_r}(n_r)) + j_r v(x) \leq v(a_{n_r}x^{n_r}).$$

The evaluated leading exponents along the chain are therefore nonincreasing, and Lemma 2.2 applies. $\square$

Note that (3.3) is a weak inequality. This is the difference between an open and a closed excluded region, and it is available only because Lemma 2.2 also rules out a constant leading exponent. Section 5.2 exhibits an equation whose excluded region is exactly the closed one, and Section 8.5 exhibits a coarsened comparison where the boundary genuinely must be left in.

Corollary 3.4 (Uniform bounded cost; no divisibility). *Suppose in (3.1) there is a single $B \in \Gamma$ with $v(c_0(n)) - v(c_j(n)) \leq B$ for all $n \geq N$ and all $j \geq 1$ with $c_j(n) \neq 0$. Put $B^+ = \max(B, 0) \in \Gamma$. Then for every $x \neq 0$ with $v(x) \leq -B^+$, strong summability of $(a_n x^n)_n$ forces $a_n = 0$ for all $n \geq N$. Such x exist: $x = t^{-B^+}$, and $x = t^{-\delta}$ for every $\delta \geq B^+$.*

Proof. Since $B^+ \geq 0$ we have $v(x) \leq 0$, so $jv(x) \leq v(x) \leq -B^+ \leq -B$ for $j \geq 1$. Hence $v(c_0(n)) - v(c_j(n)) + jv(x) \leq B - B = 0$, and Theorem 3.3 applies. Every ordered abelian group admits a maximum of finitely many elements and the monomial t^{-B^+} exists, so no cofinal cyclic subgroup is needed. $\square$

The absence of a cofinality requirement here is exactly why the mechanism proves rigidity in groups of very large rank. It is also why the refined bound below, which divides by j , is a refinement of the constant and never of the conclusion.

Corollary 3.5 (Refined periodic threshold; requires Γ divisible). *Assume in addition that Γ is divisible. Assume each slot of (3.1) is either identically zero for $n \geq N$ or nonzero for all $n \geq N$ with periodic valuation there of common period $m \geq 1$; enlarge N past any initial nonperiodic range first. Let $\alpha_{j,r}$ be the valuation of slot j on the residue class $r \bmod m$, the identically zero slots being omitted. Put*

$$B_{\text{rec}} = \max_{\substack{0 \leq r < m \\ 1 \leq j \leq s, c_j \neq 0}} \frac{\alpha_{0,r} - \alpha_{j,r}}{j} \in \Gamma. \quad (3.4)$$

If there is no active forward slot $j \geq 1$, set $B_{\text{rec}} = 0$; the recurrence then forces $a_n = 0$ for all $n \geq N$ directly. Then for every $x \neq 0$ with $v(x) \leq -B_{\text{rec}}$, strong summability of $(a_n x^n)_n$ forces $a_n = 0$ for all $n \geq N$. In particular every solution with $a_n \neq 0$ for some $n \geq N$ — so in particular every nonpolynomial solution — satisfies

$$\text{Dom}_{\text{str}}(f) \subseteq \{0\} \cup \{x \in K_\Gamma^\times : v(x) > -B_{\text{rec}}\}, \quad f = \sum_n a_n z^n. \quad (3.5)$$

Proof. The case with no forward slot was handled in the statement. Otherwise divisibility makes each quotient $(\alpha_{0,r} - \alpha_{j,r})/j$ an element of Γ , so B_{rec} is defined. If $v(x) \leq -B_{\text{rec}}$ then $jv(x) \leq -j(\alpha_{0,r} - \alpha_{j,r})/j = -(\alpha_{0,r} - \alpha_{j,r})$, because multiplying a weak inequality by the positive integer j preserves it. Hence (3.3) holds and Theorem 3.3 applies. $\square$

The inequality in (3.5) is strict on the right: at $v(x) = -B_{\text{rec}}$ the chain may keep a constant leading exponent rather than descend, which is precisely why both halves of Definition 1.2 are needed. The bound is an *exclusion* bound. It does not assert that every point on the other side lies in the domain.

4 Valuations on integer and geometric orbits

4.1 Ordinary integers cannot create unbounded valuation loss

Lemma 4.1 (Integer evaluation). *Let $0 \neq P(X) = \sum_k p_k X^k \in K_\Gamma[X]$ and put $\beta(P) = \min_{p_k \neq 0} v(p_k)$. For all but finitely many nonnegative ordinary integers n ,*

$$P(n) \neq 0 \quad \text{and} \quad v(P(n)) = \beta(P). \quad (4.1)$$

Proof. The polynomial $t^{-\beta(P)}P$ has coefficients of nonnegative valuation and at least one of valuation zero, so its coefficientwise reduction

$$\overline{P}(X) = \sum_{v(p_k) = \beta(P)} \text{lc}(p_k) X^k \in \mathbb{C}[X]$$

is nonzero and has finitely many roots. The ordinary integers have distinct images in $\mathbb{C}$. At every integer n with $\overline{P}(n) \neq 0$, the Hahn coefficient of $t^{-\beta(P)}P(n)$ at exponent zero is nonzero, and no smaller exponent occurs. $\square$

Residue characteristic zero does real work here. The proof does not apply to p -adic integer inputs, where positive integers can have arbitrarily large valuation. Conversely no Diophantine approximation over $\mathbb{C}$ is relevant to v : every nonzero complex number, however small in ordinary modulus, has valuation zero.

4.2 A geometric orbit of nonzero valuation

Lemma 4.2 (Eventually affine valuation). *Let $q \in K_\Gamma^\times$ with $\lambda = v(q) \neq 0$ and let $0 \neq P(X) = \sum_k p_k X^k \in K_\Gamma[X]$. There are an index k_P with $p_{k_P} \neq 0$, the value $\beta_P = v(p_{k_P})$, and $N_P \in \mathbb{N}$, such that*

$$v(P(q^n)) = \beta_P + k_P n \lambda \quad (n \geq N_P). \quad (4.2)$$

In particular $P(q^n)$ is eventually nonzero.

Proof. The nonzero terms have valuations $v(p_k) + kn\lambda$. For two distinct indices the difference $v(p_k) - v(p_l) + (k - l)n\lambda$ is strictly monotone in n , so its sign is eventually constant and it vanishes for at most one integer, whether or not its values are cofinal in any convex subgroup. Finitely many pairs give one common threshold beyond which every pairwise comparison is strict and fixed, hence a unique minimizing index k_P . Its term cannot cancel against terms of strictly larger valuation. $\square$

Remark 4.3 (A rank-one shortcut that fails). The minimizing index need not be the smallest when $v(q) > 0$. In $\Gamma = \mathbb{Q}H \oplus \mathbb{Q}\epsilon$ with H dominating every $n\epsilon$, take $q = t^\epsilon$ and $P(X) = X + t^{-H}$: the constant term has smaller valuation than q^n for every n , however large. Conversely with $P(X) = 1 + t^{-H}X$ the nonconstant term is the minimum for every n . The same phenomenon in the other sign convention: in $\Gamma = \mathbb{Q}e_0 \oplus \mathbb{Q}e_1$ with $e_1 \gg e_0 > 0$, $q = t^{e_0}$ and $P(X) = t^{e_1} + X$ give $v(P(q^n)) = ne_0$ for every $n \geq 0$, so the nonzero constant coefficient never becomes the least-valued term. The proof above uses the eventual ordering of finitely many affine functions, not the Archimedean assertion that $n\epsilon$ eventually exceeds H , and not the rank-one slogan that q^n eventually beats every coefficient ratio.

4.3 Valuation-zero dilations, including residues that are roots of unity

Distinct powers of a valuation-zero q all have the same valuation, so the previous lemma says nothing. Polynomial evaluation nevertheless has only a finite periodic valuation pattern.

Theorem 4.4 (Unit-orbit valuations). *Let $q \in K_\Gamma^\times$ have $v(q) = 0$ and infinite multiplicative order, and let $0 \neq P \in K_\Gamma[X]$. Then $v(P(q^n))$ is eventually finite and periodic. More precisely, with $\zeta = \text{res}(q) \in \mathbb{C}^\times$:*

- (a) *if ζ is not a root of unity, $v(P(q^n))$ is eventually constant, equal to the minimum valuation of the coefficients of P ;*
- (b) *if ζ has exact order h , then $v(P(q^n))$ is eventually constant on each residue class modulo h , so its eventual period divides h .*

Proof. For (a) write $P(X) = \sum_k p_k X^k$ and $\beta = \min_k v(p_k)$. The coefficient of $P(q^n)$ at exponent β is $\overline{P}(\zeta^n)$ with $\overline{P}(X) = \sum_{v(p_k)=\beta} \text{lc}(p_k) X^k$, a nonzero complex polynomial. The points ζ^n are distinct, so this vanishes only finitely often, and no exponent below β occurs.

For (b) write

$$q = \zeta(1 + u), \quad \eta = v(u) > 0, \quad d = \text{lc}(u).$$

Here $u = q/\zeta - 1$ has positive valuation because q/ζ has valuation zero and residue 1, and $u \neq 0$ because otherwise $q = \zeta$ would have finite order. For every positive ordinary integer n the finite binomial expansion gives

$$y_n := (1 + u)^n - 1, \quad v(y_n) = \eta, \quad \text{lc}(y_n) = nd, \quad (4.3)$$

the leading coefficient being nonzero in characteristic zero. There is no convergence assertion: for each fixed n only finitely many powers occur.

Fix $r \in \{0, \dots, h-1\}$. For $n \equiv r \pmod{h}$ we have $\zeta^n = \zeta^r$, hence $q^n = \zeta^r(1 + y_n)$. Expand the nonzero polynomial

$$P(\zeta^r(1 + Y)) = \sum_{k=0}^{\deg P} b_{r,k} Y^k \quad (4.4)$$

and set

$$\beta_r = \min_{b_{r,k} \neq 0} (v(b_{r,k}) + k\eta), \quad J_r = \{k : v(b_{r,k}) + k\eta = \beta_r\}. \quad (4.5)$$

Then $P(q^n) = \sum_k b_{r,k} y_n^k$, and its coefficient at exponent β_r is the value at n of

$$R_r(T) = \sum_{k \in J_r} \text{lc}(b_{r,k}) d^k T^k \in \mathbb{C}[T]. \quad (4.6)$$

This polynomial is nonzero: its terms have distinct degrees and all the indicated coefficients are nonzero. Outside its finitely many integer roots the coefficient at β_r survives, so $v(P(q^n)) = \beta_r$ eventually on that class. Finitely many classes give one finite exceptional range. $\square$

Remark 4.5 (A second normalization of the same proof). Both source manuscripts prove Theorem 4.4 (b), by two slightly different normalizations, and we record the other one because it is occasionally more convenient. Instead of factoring out the complex root of unity, put $q^h = 1 + u$ directly, with $\eta = v(u) > 0$ and $u \neq 0$ for the same reason. Write $n = r + hk$ and expand the polynomial $Q_r(Y) = P(q^r Y)$ as $Q_r(1 + V) = \sum_k b_{r,k} V^k$. The finite binomial expansion gives $V_k = (1 + u)^k - 1$ with $v(V_k) = \eta$ and $\text{lc}(V_k) = k \text{lc}(u)$, and $P(q^n) = Q_r(1 + V_k)$. The minimum construction (4.5) and the corresponding nonzero residue polynomial (4.6), now in the variable k rather than n , finish the argument. Neither version sums a binomial series to a noninteger exponent; every expansion is finite, and arbitrary supports in q and in the coefficients of P enter only through the leading valuation and leading coefficient of finitely many Hahn elements.

Example 4.6 (A genuine two-phase pattern). Let $q = -(1 + t^\eta)$ with $\eta > 0$. Then q is nontorsion, $\text{res}(q) = -1$ has order two, and for $P(X) = X - 1$ and $n \geq 1$

$$v(P(q^n)) = v(q^n - 1) = \begin{cases} 0, & n \text{ odd}, \\ \eta, & n \text{ even}. \end{cases}$$

For odd n the residue is -2 ; for even positive n the leading term is nt^η . The valuation sequence is periodic, not the dilation itself, and a claim of eventual *constancy* without splitting residue classes would be false. The escape argument uses the bounded valuation costs; it does not replace q by its residue.

Example 4.7 (Finite exceptional indices really occur). For $q = 1 + t^\eta$ and $P(X) = X - 1 - 7t^\eta$, the leading coefficient of $P(q^n)$ at exponent η is $n - 7$, so $v(P(q^n)) = \eta$ for positive $n \neq 7$ while $v(P(q^7)) = 2\eta$. With $P(X) = X - 1 - 3t^\eta$ the exceptional index is $n = 3$, where $P(q^3) = 3t^{2\eta} + t^{3\eta}$. The exceptions are harmless for the escape argument, since it needs only an eventual range, but they cannot be ignored in a statement asserted for all indices. This is why Lemma 4.1 and Theorem 4.4 both allow finitely many exceptions even when $\text{res}(q) = 1$.

For an ordinary complex q with $|q|_\mathbb{C} \neq 1$, the complex modulus does not alter these conclusions: as an element of the Hahn coefficient field, q still has valuation zero. The ordinary modulus and the scale valuation answer different questions.

5 Differential rigidity

5.1 From an operator to a forward recurrence

A formal series is D-finite exactly when its derivatives span a finite-dimensional space over $K_\Gamma(z)$. The D-finite/P-recursive equivalence is classical [1, Theorem 1.5]; the following records the exact finite conversion, because the valuations of the resulting recurrence, not ordinary coefficient growth, are what the proof consumes.

Lemma 5.1 (Differential coefficient recurrence). *Write*

$$L = \sum_{r=0}^R \sum_{k=0}^m c_{r,k} z^k D_z^r, \quad c_{r,k} \in K_\Gamma, \quad (5.1)$$

with at least one $c_{r,k} \neq 0$, and let $f = \sum_{n \geq 0} a_n z^n$. For $n \geq m$ the coefficient of z^n in Lf is

$$\sum_{s=-m}^R Q_s(n) a_{n+s}, \quad Q_s(X) = \sum_{r-k=s} c_{r,k} (X+s)_r, \quad (5.2)$$

where $(Y)_r = Y(Y-1)\cdots(Y-r+1)$ and $(Y)_0 = 1$. At least one Q_s is nonzero. Writing s_0 for the least active shift and $u = n + s_0$, every formal solution of $Lf = 0$ satisfies, for all sufficiently large u ,

$$\sum_{j=0}^s P_j(u) a_{u+j} = 0, \quad P_j(X) = Q_{s_0+j}(X - s_0), \quad P_0 \neq 0, \quad (5.3)$$

with $s = R - s_0$. If $s = 0$, every formal solution is a polynomial.

Proof. The operator $z^k D_z^r$ carries the term of index $n + r - k$ to degree n and multiplies it by $(n+s)_r$ with $s = r - k$, which gives (5.2). For a fixed shift s the distinct derivative orders r contribute polynomials of distinct degrees in X , so a nonempty group containing a nonzero coefficient cannot cancel identically and some Q_s is nonzero. Reindexing gives (5.3). If $s = 0$ the recurrence reads $P_0(u) a_u = 0$ with $P_0 \neq 0$, so by Lemma 4.1 all but finitely many a_u vanish. $\square$

Proposition 5.2 (Uniform obstruction for P-recursive coefficients). *Suppose (a_n) satisfies (5.3) for all $n \geq N_1$, with $P_j \in K_\Gamma[X]$ and $P_0 \neq 0$. Let $\beta_j = \beta(P_j)$ for the nonzero P_j and put*

$$B = \max(\{0\} \cup \{\beta_0 - \beta_j : j \geq 1, P_j \neq 0\}) \geq 0.$$

There is $N \geq N_1$, beyond the finitely many exceptional integers of Lemma 4.1 for the finitely many nonzero P_j , such that for every $x \neq 0$ with $v(x) \leq -B$, strong summability of $(a_n x^n)_n$ forces $a_n = 0$ for all $n \geq N$. If Γ is divisible one may replace B by the smaller threshold

$$B_{\text{rec}} = \max\left(\left\{\frac{\beta_0 - \beta_j}{j} : j \geq 1, P_j \neq 0\right\}\right), \quad (5.4)$$

which is (3.4) with period $m = 1$. If the displayed set is empty, take $B_{\text{rec}} = 0$; then every solution terminates beyond N without an evaluation hypothesis.

Proof. Discard the identically zero P_j . By Lemma 4.1 there is N beyond which $v(P_j(n)) = \beta_j$ for all the finitely many nonzero P_j simultaneously. Apply Corollary 3.4 with this B , or Corollary 3.5 with $m = 1$ when Γ is divisible. If no $j \geq 1$ remains, the recurrence forces the same vanishing directly. $\square$

These are certified exterior obstructions, not a computation of the exact evaluation domain of each solution. Individual solutions can have stricter restrictions or can terminate much earlier.

Proof of Theorem A. Lemma 5.1 converts L into (5.3). Take $B_L = B$ from Proposition 5.2 and take N_L to be an positive integer beyond the recurrence's starting index and beyond every nonnegative integer root of the finitely many residue polynomials $\overline{P_j}$. If a formal solution is strongly evaluable at one $x \neq 0$ with $v(x) \leq -B_L$, then $a_n = 0$ for all $n \geq N_L$, so its degree is less than N_L . An entire solution is in particular strongly evaluable at the available monomial t^{-B_L} , hence is a polynomial. Conversely a polynomial is entire and is annihilated by a sufficiently high derivative, which gives the displayed equality. $\square$

The closed region $v(x) \leq -B_L$, rather than an open one, comes from the weak inequality in Theorem 3.3, hence from Lemma 2.2. A polynomial solution of degree at least N_L would in any case contradict the recurrence at its last nonzero coefficient, since $P_0(n)$ no longer vanishes there.

Theorem A is stronger than merely excluding a globally extended exponential. The operator coefficients may themselves be arbitrary Hahn series carrying infinite surreal scales. There are nevertheless only finitely many of them, so their leading valuations give a finite obstruction that one additional exterior scale defeats.

5.2 An exact boundary, not only a nonexistence result

Proposition 5.3 (Formal exponential domain). *For $\eta \in \Gamma$ let*

$$E_\eta(z) = \sum_{n \geq 0} \frac{t^{n\eta}}{n!} z^n.$$

Then $D_z E_\eta = t^\eta E_\eta$ and

$$\text{Dom}_{\text{str}}(E_\eta) = \{0\} \cup \{x \in K_\Gamma^\times : v(x) > -\eta\}. \quad (5.5)$$

Proof. The differential identity is a formal coefficient comparison. Put $y = t^\eta x$ for $x \neq 0$, so the n th evaluation term is $y^n/n!$.

If $v(y) > 0$, write $y = ct^{v(y)}(1+u)$ with $c \in \mathbb{C}^\times$ and u zero or of positive valuation, and put $S = \text{supp}(u) \subseteq \Gamma_{>0}$. Since $(1+u)^n$ is a finite integer power, $\text{supp}(y^n) \subseteq h_n + M(S)$ with $h_n = nv(y)$,

and (h_n) is strictly increasing, so its values are well ordered and each is taken once. Lemma 2.3 (b) makes the family strongly summable; the factor $1/n!$ is a nonzero complex scalar and changes no support.

If $v(y) < 0$ the leading exponents $nv(y)$ strictly decrease. If $v(y) = 0$, then $y^n/n!$ has the nonzero coefficient $\text{lc}(y)^n/n!$ at exponent zero for every n , so local finiteness fails. These cases exhaust $x \neq 0$, and $v(y) > 0$ is $v(x) > -\eta$. $\square$

The recurrence of E_η is $(n+1)a_{n+1} - t^\eta a_n = 0$, with one active shift $j = 1$, so $\alpha_0 = \eta$, $\alpha_1 = 0$, and (3.4) gives $B_{\text{rec}} = \eta$ with no division by an integer at all. Comparing with (5.5): the excluded region $v(x) \leq -\eta$ of Corollary 3.5 is here *exactly* the complement of the domain, boundary included. The general exclusion bound can therefore be attained, and the closed form of the boundary is not an artefact of a lossy estimate. This example is the reason for keeping Lemma 2.2 rather than the strictly descending version alone.

There is no conflict with any scalar exponential on a surreal field: E_η is an ordinary power series in z , not a total exponential evaluated by an independent construction. Taking $\eta = 0$ recovers the elementary observation (2.3).

5.3 Algebraic and system consequences

Corollary 5.4 (Algebraic entire series). *An element of $\mathcal{E}_\Gamma$ algebraic over $K_\Gamma(z)$ belongs to $K_\Gamma[z]$.*

Proof. Let $F = K_\Gamma(z, f)$ be the finite extension generated by the algebraic series. In characteristic zero its minimal polynomial is separable, and differentiating that polynomial expresses $D_z f$ inside F , because the derivative with respect to the algebraic variable is nonzero at f and hence invertible in F . So D_z preserves F , all derivatives of f lie in the finite-dimensional $K_\Gamma(z)$ -space F , and f is D-finite. Apply Theorem A. $\square$

This is not offered as a new algebraic rigidity theorem; it records that the linear result subsumes the algebraic case without any factorization theorem for $\mathcal{E}_\Gamma$.

Corollary 5.5 (Rational differential systems). *If a finite vector Y of strongly entire formal series satisfies $D_z Y = A(z)Y$ with $A \in M_r(K_\Gamma(z))$, every component of Y is a polynomial. The same holds for an inhomogeneous system with a rational forcing vector, provided all its solution components are strongly entire.*

Proof. The $K_\Gamma(z)$ -span of the components is stable under D_z , so the successive derivatives of each component span a finite-dimensional space. For $Y' = AY + b$ append the constant component 1 and use $\begin{pmatrix} A & b \\ 0 & 0 \end{pmatrix}$. $\square$

Proposition 5.6 (Rational inhomogeneous equations). *The polynomiality conclusions of this report remain valid when the right-hand side of a linear equation is a rational function of z , provided the corresponding homogeneous hypothesis on L or on q holds.*

Proof. Clear denominators in the coefficients and in the right-hand side. The result has a polynomial right-hand side, so the coefficient recurrence is homogeneous for all sufficiently large indices, and the escape statements were formulated for eventual recurrences. The left-hand operator is still nonzero, having been multiplied by a nonzero polynomial. $\square$

This applies, for instance, to the inhomogeneous first-order theta identity (8.2) when the parameter valuation is noncofinal: it is not necessary to homogenize it in order to see the obstruction.

Example 5.7 (A sparse recurrence does not escape the obstruction). The formal Airy-type equation $f'' - zf = 0$ has the recurrence

$$a_{n+3} = \frac{a_n}{(n+3)(n+2)} \quad (n \geq 0), \quad a_2 = 0.$$

With $a_0 = 1$ and $a_1 = 0$ there are infinitely many nonzero coefficients, at indices $3n$, all of valuation zero. At $x = t^{-\delta}$ their leading exponents are $-3n\delta$, strictly decreasing for every $\delta > 0$. Gaps between nonzero coefficients do not help: the escape chain follows only the surviving congruence class.

Remark 5.8 (What transcendence has actually been proved). Every nonpolynomial member of $\mathcal{E}_\Gamma$ is transcendental over $K_\Gamma(z)$ and satisfies no nonzero *linear* differential equation over $K_\Gamma(z)$, and no finite family $f, D_z f, \dots, D_z^R f$ is linearly dependent over $K_\Gamma(z)$. This does not imply differential transcendence in the nonlinear sense. A relation

$$P(z, f, D_z f, \dots, D_z^R f) = 0$$

involving products of f and its derivatives lies outside the present proofs unless it can separately be reduced to a finite linear system. The distinction is not cosmetic: the escape mechanism exploits a finite *linear* cancellation in which one nonzero coefficient forces a later nonzero coefficient, whereas nonlinear coefficient equations involve convolutions of many earlier coefficients and supply no such forward step without further work.

6 Generic lines and several-variable rigidity

A single differential equation in several variables is not enough to force polynomiality: it could ignore most of the variables. The correct hypothesis is finite-dimensionality of the span of *all* mixed partial derivatives, which is what Theorem C assumes.

Lemma 6.1 (Avoiding countably many polynomial conditions). *For any countable family of nonzero polynomials in $K_\Gamma[z_1, \dots, z_d]$ there is $c \in \mathbb{C}^d$ at which none of them vanishes.*

Proof. For one polynomial H , take the least Hahn exponent among its finitely many nonzero coefficients; the coefficient at that exponent is a nonzero $\overline{H} \in \mathbb{C}[z_1, \dots, z_d]$, and $\overline{H}(c) \neq 0$ implies $H(c) \neq 0$. The coefficients of the countably many polynomials $\overline{H}$ form a countable subset of $\mathbb{C}$, and the field they generate over $\mathbb{Q}$ is countable. Choose $c_1, \dots, c_d$ algebraically independent over that field, successively outside the algebraic closure of a countable field, which remains countable while $\mathbb{C}$ is uncountable. $\square$

The same countable-field device is used in the companion report held in `docs/surcomplex/entire-functions-at-arbitrary-rank/` (`ent:mv:thm:exceptional`) to show that the set of polynomial directions of a nonpolynomial series is not all of projective space.

Proof of Theorem C. Suppose F is not a polynomial and write it by total degree,

$$F(z) = \sum_{n \geq 0} F_n(z), \quad F_n \text{ homogeneous of degree } n, \quad (6.1)$$

so infinitely many F_n are nonzero. Let $g_1, \dots, g_\sigma$ be a basis of the finite-dimensional $K_\Gamma(z_1, \dots, z_d)$ -span of all mixed derivatives of F , taken to be mixed derivatives themselves with $g_1 = F$. The span is stable under differentiation, so there are rational matrices A_i with

$$\partial_i g = A_i(z)g \quad (1 \leq i \leq d). \quad (6.2)$$

These matrices have finitely many nonzero denominator polynomials; choose one nonzero homogeneous component of each. Apply Lemma 6.1 to every nonzero F_n , to those finitely many chosen components, and to the coordinate polynomials z_i , obtaining $c \in \mathbb{C}^d$ with $F_n(c) \neq 0$ whenever $F_n \neq 0$ and with no denominator becoming the zero polynomial under $z = wc$. Then

$$h(w) = F(wc) = \sum_{n \geq 0} F_n(c)w^n$$

is nonpolynomial.

For each $x \in K_\Gamma$ the family $(a_\alpha c^\alpha x^{|\alpha|})_{\alpha \in \mathbb{N}^d}$ is strongly summable, by entireness of F at $(c_1 x, \dots, c_d x)$. Group it into the finite blocks of fixed total degree; by the last clause of Lemma 2.3, $\sum_n F_n(c)x^n$ is strongly summable, so $h \in \mathcal{E}_\Gamma$.

The formal chain rule, applied after clearing denominators in (6.2), gives

$$\frac{d}{dw}g(wc) = \left(\sum_{i=1}^d c_i A_i(wc) \right) g(wc). \quad (6.3)$$

Every denominator here is a nonzero element of $K_\Gamma[w]$. It may vanish at $w = 0$, which is harmless: read the identity in the Laurent series field $K_\Gamma((w))$, with matrices in $M_\sigma(K_\Gamma(w))$. The finite $K_\Gamma(w)$ -span of the components of $g(wc)$ is preserved by d/dw , so its first component h is D-finite over $K_\Gamma(w)$, and Theorem A forces h to be a polynomial. This contradicts its construction. $\square$

Corollary 6.2 (Multivariate algebraic rigidity). *If a strongly entire $F \in K_\Gamma[[z_1, \dots, z_d]]$ is algebraic over $K_\Gamma(z_1, \dots, z_d)$, it is a polynomial.*

Proof. Each partial derivation extends to the finite separable extension generated by F , by differentiation of its minimal polynomial, so all mixed partial derivatives lie in one finite-dimensional space. Apply Theorem C. $\square$

The generic line is an existence device, not a promised effective choice of ordinary complex constants. The proof does not say that one prescribed line detects every nonpolynomial multivariate series; the line is chosen only after the countably many nonzero homogeneous parts and the finitely many rational connection matrices are known.

7 Dilation equations and the necessity of a cofinal scale

7.1 Exact coefficient conversion

Lemma 7.1 (Dilation coefficient recurrence). *Let $q \in K_\Gamma^\times$ have infinite multiplicative order and let $f = \sum a_n z^n$ satisfy (1.3). Write $p_\ell(z) = \sum_k p_{\ell,k} z^k$ and $Q_k(X) = \sum_\ell p_{\ell,k} X^\ell$, let d and e be the largest and smallest k with $Q_k \neq 0$, and put $s = d - e$. Then for every $n \geq 0$*

$$\sum_{h=0}^s R_h(q^n) a_{n+h} = 0, \quad R_h(X) = Q_{d-h}(q^h X), \quad (7.1)$$

with $R_0 = Q_d \neq 0$ and $R_s = Q_e(q^s \cdot) \neq 0$. Each nonzero R_h satisfies $R_h(q^n) \neq 0$ for all but finitely many n . If $s = 0$, every formal solution is a polynomial.

Proof. The coefficient of z^{n+d} in $\sum_\ell p_\ell(z)f(q^\ell z)$ is $\sum_{k=e}^d Q_k(q^{n+d-k})a_{n+d-k}$; substituting $h = d - k$ and using $Q_{d-h}(q^{n+h}) = Q_{d-h}(q^h \cdot q^n)$ gives (7.1). Since q has infinite order the points q^n are distinct, and a nonzero univariate polynomial vanishes at only finitely many of them; this already gives eventual nonvanishing, independently of the finer valuation results. If $s = 0$ the relation reads $R_0(q^n)a_n = 0$ for every n , so all but finitely many a_n vanish. $\square$

The powers q^h in (7.1) matter: dropping them changes the recurrence and can change its valuation constants. Equivalently, taking $m = \max_\ell \deg p_\ell$ and reading off the coefficient of z^{n+m} gives $\sum_j P_j(q^n)a_{n+j} = 0$ with $P_j(X) = \sum_\ell p_{\ell, m-j} q^{\ell j} X^\ell$ and $P_0 = Q_m \neq 0$; both conversions are checked on finite exact examples by the programs of Section 14. If $m = 0$ the equation already reads $P_0(q^n)a_n = 0$ and all solutions are polynomials with no entireness assumption at all, so we may assume $s \geq 1$ below.

7.2 The valuation-zero case

Corollary 7.2 (Every unit dilation is globally rigid). *Let $q \in K_\Gamma^\times$ have $v(q) = 0$ and infinite multiplicative order. Every linearly q -difference finite member of $\mathcal{E}_\Gamma$ is a polynomial, and each fixed equation (1.3) has a uniform exterior obstruction and a uniform degree bound. If Γ is in addition divisible, every nonpolynomial formal solution has its domain bounded as in (3.5) by the constant (3.4) formed from the eventual residue-class valuations of (7.1).*

Proof. Apply Theorem 4.4 simultaneously to the finitely many nonzero R_h of Lemma 7.1. Their valuations take only finitely many values after a common starting index, with common period one or the exact order h_0 of $\text{res}(q)$. All differences $v(R_0(q^n)) - v(R_h(q^n))$ are therefore bounded above by a single element of Γ , and Corollary 3.4 gives the exterior obstruction and, via the residue polynomials, the degree bound. When Γ is divisible, Corollary 3.5 applies with $m = 1$ if the residue has infinite order, and $m = h_0$ otherwise. In both cases take the starting index beyond the exceptional range. $\square$

This includes parameters arbitrarily close to a root of unity without being one. The finite exceptional cancellations of Theorem 4.4 can move the starting index but do not remove the bound.

7.3 Nonzero but noncofinal valuation

Proposition 7.3 (Noncofinal dilation scales force polynomiality). *Let $q \in K_\Gamma^\times$ with $\beta = v(q) \neq 0$ and suppose $|\beta|$ is not an order unit of Γ . Every linearly q -difference finite member of $\mathcal{E}_\Gamma$ is a polynomial, with a uniform exterior obstruction and a uniform degree bound for each fixed equation. Divisibility of Γ is not used.*

Proof. By Lemma 4.2 applied to the finitely many nonzero R_h of (7.1), there is a common index beyond which

$$v(R_h(q^n)) = \alpha_h + k_h n \beta, \quad k_h \in \mathbb{N}.$$

Put $V = |\beta| > 0$, choose $B \in \Gamma$ with $B \geq 0$ and $B \geq |\alpha_0 - \alpha_h|$ for all relevant h , and put $M = \max_h |k_0 - k_h| \in \mathbb{N}$. Then for $h \geq 1$

$$v(R_0(q^n)) - v(R_h(q^n)) = \alpha_0 - \alpha_h + (k_0 - k_h)n\beta \leq B + MnV. \quad (7.2)$$

Because $|\beta|$ is not an order unit, (2.2) supplies $\gamma > 0$ with $\gamma > NV$ for every $N \in \mathbb{N}$; in particular $MnV < \gamma$ for all n , trivially so if $M = 0$. Hence the right side of (7.2) is bounded above by the single element $B + \gamma$. Corollary 3.4 with this bound excludes every $x \neq 0$ with $v(x) \leq -(B + \gamma)$, and the explicit witness $x = t^{-(B+\gamma)}$ is available. Since an entire solution is evaluable there, it satisfies $a_n = 0$ beyond the common starting index, hence is a polynomial of degree less than that index. Nothing in this argument divides an element of Γ by an integer. $\square$

The proof covers two different higher-rank situations. The group may have no order unit at all, so every nonzero dilation valuation is noncofinal; alternatively the group may have an order unit while the chosen q lives on a smaller scale. Global q -difference finiteness depends on the valuation of that particular dilation, not only on the group. The argument also requires no asymptotic formula for $v(a_n)$ and no eventually quadratic coefficient valuation: a single infinite escape chain suffices, which is why cancellations and sparse zero coefficients do not force a separate Skolem–Mahler–Lech analysis.

7.4 An intrinsic coarsened exclusion bound

The previous proof produces more than one bad point. Let $\beta = v(q) \neq 0$, let $\Delta = \Delta_{|\beta|}$ be the convex subgroup (2.1), and assume $\Delta \neq \Gamma$. Let $\pi : \Gamma \rightarrow \Gamma/\Delta$ be the ordered quotient and put $w = \pi \circ v$. With the eventual constants α_h of the previous proof, put

$$\overline{B}_{\text{rec}}^b = \max(\{0\} \cup \{\pi(\alpha_0) - \pi(\alpha_h) : 1 \leq h \leq s, R_h \neq 0\}), \quad (7.3)$$

and, when Γ is divisible so that Γ/Δ is a divisible ordered group,

$$\overline{B}_{\text{rec}} = \max_{1 \leq h \leq s, R_h \neq 0} \frac{\pi(\alpha_0) - \pi(\alpha_h)}{h}. \quad (7.4)$$

Theorem 7.4 (Coarsened exclusion). *Let $f = \sum a_n z^n$ be a nonpolynomial formal solution of (1.3) with $v(q) \neq 0$ and $\Delta = \Delta_{|v(q)|}$ proper.*

- (a) *Under Convention 1.1 alone, every $x \neq 0$ with $-w(x) > \overline{B}_{\text{rec}}^b$ lies outside $\text{Dom}_{\text{str}}(f)$.*
- (b) *If Γ is in addition divisible, every $x \neq 0$ with $-w(x) > \overline{B}_{\text{rec}}$ lies outside $\text{Dom}_{\text{str}}(f)$.*

Proof. Coarsening by Δ kills $n\beta$, since $n\beta \in \Delta$ for every n , so the projected valuations of the recurrence coefficients are the constants $\pi(\alpha_h)$. In case (b), $-w(x) > \overline{B}_{\text{rec}}$ gives $hw(x) < -(\pi(\alpha_0) - \pi(\alpha_h))$ for every relevant $h \geq 1$. In case (a) we have $w(x) < -\overline{B}_{\text{rec}}^b \leq 0$, hence $hw(x) \leq w(x) < -\overline{B}_{\text{rec}}^b \leq -(\pi(\alpha_0) - \pi(\alpha_h))$. Either way the projected leading valuation along an escape chain strictly decreases at every step. A strict decrease in the ordered quotient is a strict decrease in Γ , so the chain produces an infinite strictly descending sequence in the union of the supports. $\square$

The boundary is deliberately *not* excluded here. Equal leading values in the quotient can correspond to a strictly increasing sequence in Γ , and hence to a strongly summable family; Section 8.5 exhibits exactly that. Unlike Corollary 3.5, a coarsened comparison alone cannot decide local finiteness at an exponent of Γ .

8 Partial theta series: sharp witness and exact domain

8.1 The classical identities

For $p \in K_\Gamma^\times$ with $\beta = v(p) > 0$ set

$$\Theta_p(z) = \sum_{n \geq 0} p^{\binom{n}{2}} z^n = \sum_{n \geq 0} p^{n(n-1)/2} z^n. \quad (8.1)$$

This is a standard partial theta series up to a harmless sign convention [5]. None of its coefficients vanishes, so it is not a polynomial. The coefficient identity $p^{n(n-1)/2} = p^{n-1} p^{(n-1)(n-2)/2}$ for $n \geq 1$ gives

$$\Theta_p(z) = 1 + z\Theta_p(pz), \quad (8.2)$$

and applying $\sigma_p - 1$ to $(1 - z\sigma_p)\Theta_p = 1$, with $\sigma_p z = pz\sigma_p$, yields the homogeneous equation

$$\Theta_p(z) - (1 + z)\Theta_p(pz) + pz\Theta_p(p^2z) = 0, \quad (8.3)$$

which is also obtained by substituting pz for z in (8.2) and subtracting the two inhomogeneous identities. The existence of these identities is not the new part. What matters is precisely where the series is strongly evaluable, for an arbitrary-rank value group and an arbitrary Hahn parameter.

8.2 Exact domain with arbitrary coefficient tails

Theorem 8.1 (Exact partial theta domain). *Let Γ be as in Convention 1.1 and let $p \in K_\Gamma^\times$ with $\beta = v(p) > 0$, and let Δ_β be as in (2.1). Then*

$$\begin{aligned} \text{Dom}_{\text{str}}(\Theta_p) &= \{0\} \cup \{x \in K_\Gamma^\times : v(x) \geq -N\beta \text{ for some } N \in \mathbb{N}\} \\ &= \{0\} \cup \{x \in K_\Gamma^\times : (v \bmod \Delta_\beta)(x) \geq 0\}. \end{aligned} \quad (8.4)$$

In particular $\Theta_p \in \mathcal{E}_\Gamma$ if and only if β is an order unit of Γ . No monomial hypothesis on p or on x is needed, and Γ need not be divisible.

Proof. Evaluation at 0 is finite. For $x \neq 0$ put $\delta = v(x)$ and write

$$p = ct^\beta(1+u), \quad x = dt^\delta(1+w), \quad c, d \in \mathbb{C}^\times,$$

where each of u, w is zero or has strictly positive valuation. Let $S = \text{supp}(u) \cup \text{supp}(w) \subseteq \Gamma_{>0}$ and $M = M(S)$. The n th evaluation term is

$$p^{\binom{n}{2}} x^n = c^{\binom{n}{2}} d^n t^{h_n} (1+u)^{\binom{n}{2}} (1+w)^n, \quad h_n = \binom{n}{2} \beta + n\delta,$$

and the last two factors are finite integer powers, so

$$\text{supp}(p^{\binom{n}{2}} x^n) \subseteq h_n + M, \quad v(p^{\binom{n}{2}} x^n) = h_n. \quad (8.5)$$

Suppose first that $\delta \geq -N\beta$ for some $N \in \mathbb{N}$. Then

$$h_{n+1} - h_n = n\beta + \delta \geq (n - N)\beta > 0 \quad (n > N),$$

so (h_n) is eventually strictly increasing. Its set of values is therefore well ordered, including the finite initial segment, and each value is taken by finitely many indices. Lemma 2.3(b) applied to (8.5) proves strong summability.

If no such N exists, then $\delta < -N\beta$ for every N , so $h_{n+1} - h_n = n\beta + \delta < 0$ for every n . By (8.5) the leading exponents are exactly h_n , and they form a strictly decreasing sequence, so strong summability fails by Lemma 2.2.

Finally, $\delta \geq -N\beta$ for some N says exactly that the image of δ in Γ/Δ_β is nonnegative: a nonnegative coset either is positive, whence $\delta > 0$, or is zero, whence $\delta \in \Delta_\beta$ and $\delta \geq -N\beta$ for some N ; a negative coset lies below every element of Δ_β . The last assertion is the case $\Delta_\beta = \Gamma$. $\square$

This proof controls the whole support, not only the leading valuation. Leading valuation formulas by themselves do not in general determine a Hahn evaluation domain, because hidden tails can matter; here the common positive monoid M is the extra certificate that makes the formula exact. Note also that the entire calculation is carried out in Γ : nothing is divided by an integer, and no real-valued surrogate for v is introduced.

Remark 8.2 (A second, shorter proof of the order-unit half). When $\mu = \beta$ is an order unit, entireness also follows from the cofinal-valuation criterion Lemma 2.3(a) alone, and both source manuscripts are kept here because the two routes are genuinely different. For $x \neq 0$ with $\alpha = v(x)$, the n th evaluated term has valuation

$$\frac{n(n-1)}{2}\mu + n\alpha. \quad (8.6)$$

Given a target $\beta' \in \Gamma$, choose ordinary nonnegative integers M', B' with $\alpha \geq -M'\mu$ and $\beta' \leq B'\mu$, which is possible exactly because μ is an order unit. Then (8.6) is at least $(\frac{n(n-1)}{2} - M'n)\mu$, which exceeds $B'\mu$ for all sufficiently large ordinary n . So the term valuations tend cofinally to $+\infty$ and Lemma 2.3(a) applies. Only leading term valuations are used, so unrestricted higher Hahn tails in p do not invalidate the conclusion. This shorter route is available only in the order-unit case: at the boundary $v(x) = -N\beta$ with β noncofinal, the term valuations increase strictly but stay trapped below a fixed element of Γ , so criterion (a) says nothing and the monoid certificate is essential.

Remark 8.3 (The order-unit case has an optional rank-one picture). When β is an order unit, the real-valued coarsening $\pi : \Gamma \rightarrow \mathbb{R}$ of [12, Lemma 9.1], normalized by $\pi(\beta) = 1$, gives the same field K_Γ a rank-one absolute value. For $x \neq 0$, its theta terms have coarsened valuations $\binom{n}{2} + n\pi(v(x))$, tending to $+\infty$. This is an optional normed-field picture of the order-unit case, not a map that replaces the Hahn exponents by their images in $\mathbb{R}$.

Such a coefficientwise replacement need not define a Hahn series. In $\mathbb{Q}H \oplus \mathbb{Q}\epsilon$ with H dominating all $n\epsilon$, the coarsening normalized by H kills ϵ . The valid series $\sum_{n \geq 0} t^{n\epsilon}$ would then place infinitely many coefficients at exponent zero. Coarsening a valuation retains the field and changes the value map; it is not an embedding into a complex-coefficient Hahn field with smaller exponent group. No proof here requires this optional coarsening.

8.3 Completion of the sharp criterion

Proof of Theorem B. Assume (i) and suppose (ii) fails. If $v(q) = 0$, Corollary 7.2 makes every entire solution a polynomial. If $v(q) \neq 0$ and $|v(q)|$ is not an order unit, Proposition 7.3 does the same. Either way (i) fails, which proves (i) $\Rightarrow$ (ii) and supplies the stated uniform obstruction and degree bound.

Assume (ii). If $v(q) > 0$ is an order unit take $p = q$: by Theorem 8.1, $\Theta_q \in \mathcal{E}_\Gamma \setminus K_\Gamma[z]$, and (8.3) is a nonzero linear q -difference equation with polynomial coefficients. If $v(q) < 0$ and $-v(q)$ is an order unit take $p = q^{-1}$, so $v(p) = -v(q) > 0$ is an order unit and $\Theta_{q^{-1}} \in \mathcal{E}_\Gamma$ by the same theorem. Substituting $q^2 z$ for z in (8.3), with $p = q^{-1}$ so that $pz \mapsto qz$ and $p^2 z \mapsto z$, gives

$$\Theta_{q^{-1}}(q^2 z) - (1 + q^2 z)\Theta_{q^{-1}}(qz) + qz \Theta_{q^{-1}}(z) = 0, \quad (8.7)$$

which uses only nonnegative powers of σ_q and has polynomial coefficients that are not all zero. Both signs of $v(q)$ are covered. $\square$

The partial theta function is used as an established type of construction, not advertised as a newly discovered special function. Its role is precise: it realizes the positive side of a necessary and sufficient criterion, including nonmonomial q and arbitrary valuation rank.

8.4 Two scales in the same rank-two field

Example 8.4 (One field, opposite answers). Let $\Gamma = \mathbb{Q}e_0 \oplus \mathbb{Q}e_1$ with $e_1 \gg e_0 > 0$, ordered by the highest nonzero coordinate. Then e_1 is an order unit and e_0 is not.

For $q = t^{e_0}$, no nonpolynomial strongly entire series satisfies a nonzero linear q -difference equation. The series Θ_q itself is not entire: by Theorem 8.1 its exact domain is the valuation ring obtained by forgetting the e_0 coordinate, which contains every x with $v(x) = -Ne_0$ and excludes $x = t^{-e_1}$.

For $q = t^{e_1}$, Θ_q is strongly entire, though the value group still has rank two. The two conclusions occur in the *same* Hahn field: the distinction is the dilation scale, not the rank of the field.

Under $e_0 = 1$, $e_1 = \omega$ and $t^\gamma = \omega^{-\gamma}$, the first dilation is multiplication by ω^{-1} and the second by $\omega^{-\omega}$. Quadratic multiples of the first valuation never reach the second scale; quadratic multiples of the second dominate every scale in this workspace. A fuller table for the same group, including unit dilations and the torsion exception, appears in Section 11.3.

8.5 The coarse boundary really can be included

For the theta series the forward recurrence is $a_{n+1} = p^n a_n$, read off from (8.2), whose inhomogeneity affects only the index $n = 0$ and is removed by Proposition 5.6. Its two slots have valuations $n\beta$ and 0, both of which project to 0 modulo Δ_β , so the projected gap is zero and $\overline{B}_{\text{rec}} = \overline{B}_{\text{rec}}^\flat = 0$. Theorem 7.4 therefore excludes the points of negative coarsened valuation, while Theorem 8.1 includes *all* points of zero coarsened valuation. The strict inequality in the coarsened exclusion theorem cannot be replaced by a weak one in general. This is the exact mirror image of Section 5.2, where the uncoarsened bound was attained with its boundary excluded from the domain. The two boundaries behave differently because one comparison is made in Γ and the other in a quotient that has forgotten the scale doing the work.

9 Torsion parameters and why they are excluded

If $q^h = 1$ then $\sigma_q^h - 1$ annihilates *every* formal power series, so the unrestricted definition of linear q -difference finiteness is vacuous for such a parameter. This is not a technical inconvenience removable by a better proof; it is a genuine counterexample to dropping the nontorsion hypothesis in Theorem B.

The difficulty persists even after excluding operators with zero action. For $h > 1$ and

$$f(z) = z^r g(z^h),$$

one has $f(qz) = q^r f(z)$, and the operator $\sigma_q - q^r$ is not zero on all formal series when $h > 1$. A nonpolynomial entire g produces a nonpolynomial entire f , since evaluation of f at x is evaluation of g at x^h followed by multiplication by x^r . Finite-order dilations therefore have a real periodic-support exception and are not a limiting case of Theorem B. In particular “the residue of q is a root of unity” and “ q is a root of unity” are fundamentally different hypotheses: Example 4.6 is covered by the theorem and this section is not.

There is nevertheless a meaningful eigenspace statement.

Proposition 9.1 (Torsion covariance). *Let $q \in \mathbb{C}^\times$ have exact finite order m and let $0 \leq r < m$.*

- (a) *A formal series satisfies $f(qz) = q^r f(z)$ if and only if $f(z) = z^r G(z^m)$ for a unique $G \in K_\Gamma[[z]]$.*
- (b) *If $G \in \mathcal{E}_\Gamma$ then $f = z^r G(z^m) \in \mathcal{E}_\Gamma$.*
- (c) *The converse of (b) also holds, so $f \in \mathcal{E}_\Gamma$ if and only if $G \in \mathcal{E}_\Gamma$, without divisibility of Γ .*

Moreover, every root of unity of K_Γ lies in $\mathbb{C}$, so the hypothesis $q \in \mathbb{C}^\times$ loses no torsion element of the Hahn field. All parts use only Convention 1.1.

Proof. For (a), coefficient comparison gives $(q^n - q^r)a_n = 0$, so only degrees congruent to r modulo m occur, and G is read off uniquely.

For (b), evaluation of $z^r G(z^m)$ at x is x^r times the family for G at x^m , reindexed; multiplication by the fixed element x^r preserves strong summability by Lemma 2.1 (3), and at $x = 0$ the evaluation is finite.

For (c), suppose $f \in \mathcal{E}_\Gamma$ and let $y \in K_\Gamma^\times$. Set $\delta = v(y)$, $\gamma = \min(\delta, 0)$, and $x = t^\gamma$. Since $m \geq 1$, we have $v(x^m) = m\gamma \leq \delta$. Strong evaluation of f at x , restricted to the degrees $r + mn$ and multiplied by x^{-r} , gives strong evaluation of G at x^m . Lemma 2.4 then gives strong evaluation at y . At $y = 0$ evaluation is finite. No root of y is extracted.

Finally, let $\zeta \in K_\Gamma$ be a root of unity. Then $v(\zeta) = 0$; dividing by $\text{res}(\zeta)$ gives $1 + u$ with u zero or of positive valuation, and $(1 + u)^{m'} = 1$ for some $m' \geq 1$. If $u \neq 0$ then $(1 + u)^{m'} - 1$ has valuation $v(u)$ and leading coefficient $m' \text{lc}(u) \neq 0$ in characteristic zero, a contradiction. So $u = 0$ and $\zeta = \text{res}(\zeta) \in \mathbb{C}$. $\square$

The converse uses only that the values $m\gamma$ are coinital in Γ , as the explicit choice $\gamma = \min(\delta, 0)$ shows. Actual m th roots need not exist in the original Hahn field; this stronger algebraic requirement is unnecessary for the entireness equivalence.

10 An extension to mixed differential–dilation equations

For a nonzero noncofinal dilation valuation, derivatives do not repair the missing-scale problem. The extension is useful when a symbolic system allows both differentiation and rescaling of the argument.

Lemma 10.1 (Polynomial dependence on the index as well as the dilation). *Let $q \in K_\Gamma^\times$ with $\lambda = v(q) \neq 0$ and let $0 \neq P(U, X) \in K_\Gamma[U, X]$. There are $\beta \in \Gamma$, $k \in \mathbb{N}$ and $N \in \mathbb{N}$ with*

$$v(P(n, q^n)) = \beta + kn\lambda \quad (n \geq N).$$

Proof. Write $P(U, X) = \sum_k C_k(U)X^k$. By Lemma 4.1 each nonzero $C_k(n)$ has the fixed valuation $\beta_k = \beta(C_k)$ for all sufficiently large n . The finitely many summands then have valuations $\beta_k + kn\lambda$, and the pairwise-order argument of Lemma 4.2 produces a unique eventual minimizing index, whose term cannot cancel. $\square$

Theorem 10.2 (Mixed rigidity at a noncofinal nonzero scale). *Let Γ be as in Convention 1.1, let $q \in K_\Gamma^\times$ with $v(q) \neq 0$, and suppose $|v(q)|$ is not an order unit of Γ . If a strongly entire f satisfies a nonzero equation*

$$\sum_{r,\ell} p_{r,\ell}(z) D_z^r \sigma_q^\ell f(z) = 0, \quad p_{r,\ell} \in K_\Gamma[z], \quad (10.1)$$

where the sum is finite and the coefficients are not all zero, then f is a polynomial. Each fixed equation has a uniform exterior obstruction and a uniform degree bound.

Proof. Expand $p_{r,\ell}(z) = \sum_k c_{r,\ell,k} z^k$. With $s = r - k$, the term of coefficient index $n + s$ contributes to degree n the quantity $c_{r,\ell,k}(n + s) \underline{r} q^{\ell(n+s)} a_{n+s}$, so the coefficient recurrence has shift polynomials

$$P_s(U, X) = \sum_{r-k=s} \sum_{\ell} c_{r,\ell,k} q^{\ell s} (U + s) \underline{r} X^\ell, \quad (10.2)$$

evaluated at $(U, X) = (n, q^n)$. For fixed s and ℓ , distinct r give distinct degrees in U , so a nonempty group of nonzero coefficients cannot cancel identically and some P_s is nonzero. Reindex from the smallest active shift; replacing U by $U - s_0$ and X by $q^{-s_0} X$ preserves polynomiality and nonzeroness, so the leftmost coefficient is $A_0(n) = P'_0(n, q^n)$ for a nonzero bivariate P'_0 , and Lemma 10.1 makes it eventually nonzero.

Lemma 10.1 gives eventual affine valuations for all the nonzero coefficients, so their differences obey a bound $B + Mn|v(q)|$ exactly as in (7.2). Since $|v(q)|$ is not an order unit, (2.2) bounds this by one element of Γ , and Corollary 3.4 finishes the proof. $\square$

The hypothesis $v(q) \neq 0$ is essential to *this* proof. For $v(q) = 0$ the separate pure differential and pure dilation conclusions have been proved above, but they are not a proof of the mixed conclusion, and no mixed unit-valuation classification is asserted anywhere in this report; see Question 15.2.

11 Surreal workspaces that separate the theories

The results so far are stated in set-sized Hahn fields, which keeps the foundations transparent and allows a literal surreal reading without quantifying over a proper class as though it were a set.

11.1 The workspace without an order unit

Take

$$\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q} e_j, \quad 0 < e_0 \ll e_1 \ll e_2 \ll \cdots, \quad (11.1)$$

ordered so that the highest index with a nonzero coordinate determines the sign, equivalently $e_{j+1} > N e_j > 0$ for all $j \geq 0$ and $N \geq 1$. Every element has only finitely many nonzero coordinates. One may identify e_j with the surreal ω^j .

The group has no order unit: if $\gamma > 0$ has largest active index j , then $e_{j+1} > n\gamma$ for every ordinary n . The sequence (e_n) is nevertheless cofinal in the whole group, so this is a countable

cofinality that cannot be generated by repeatedly adding one positive element. This group and this distinction appear in the companion entire-function report [12] and are background here, not a new observation.

Corollary 11.1 (Universal nontorsion dilation rigidity). *Suppose Γ has no order unit. Then every nonpolynomial $f \in \mathcal{E}_\Gamma$ satisfies, simultaneously for all nontorsion $q \in K_\Gamma^\times$ and all $r \geq 0$,*

$$f(z), f(qz), \dots, f(q^r z) \quad \text{are linearly independent over } K_\Gamma(z).$$

Such fields can contain nonpolynomial strongly entire series, as (11.1) and Theorem 11.2 show.

Proof. A linear dependence over $K_\Gamma(z)$ becomes, after clearing denominators, a nonzero equation (1.3). Since Γ has no order unit, no $|v(q)|$ is an order unit, so Theorem B forces f to be a polynomial. $\square$

Theorem 11.2 (An explicit universally nonholonomic entire series). *In K_{Γ_∞} the series*

$$F(z) = \sum_{n \geq 0} t^{e_n} z^n = \sum_{n \geq 0} \omega^{-\omega^n} z^n \tag{11.2}$$

is strongly entire and nonpolynomial. It satisfies no nonzero linear differential equation over $K_{\Gamma_\infty}(z)$ and, for every nontorsion $q \in K_{\Gamma_\infty}^\times$, no nonzero polynomial-coefficient linear q -difference equation. For every q of nonzero valuation it satisfies no nonzero mixed equation (10.1) either.

Proof. Fix $x \neq 0$ and write $x = ct^\delta(1+w)$ with $c \in \mathbb{C}^\times$ and w zero or of positive valuation. Choose M with δ involving only the coordinates $e_0, \dots, e_M$. The n th evaluation term has leading exponent $h_n = e_n + n\delta$ and support contained in $h_n + M(\text{supp } w)$. For $n > M$ the highest coordinate of h_n is e_n with coefficient 1, so $h_n > 0$, and the highest coordinate of

$$h_{n+1} - h_n = e_{n+1} - e_n + \delta$$

is e_{n+1} with coefficient 1, so (h_n) is eventually strictly increasing. Lemma 2.3 (b) proves strong summability; at $x = 0$ only the constant term remains. All coefficients are nonzero, so F is nonpolynomial.

Alternatively, for any $\beta \in \Gamma_\infty$ choose j larger than every active coordinate index of δ and of β ; for $n > j$ the largest active coordinate of $h_n - \beta$ is n with coefficient 1, so $h_n > \beta$, and Lemma 2.3 (a) applies for arbitrary w . Cofinal leading valuations control the whole support, since every exponent of a Hahn term is at least its leading exponent.

The differential conclusion is Theorem A. Since Γ_∞ has no order unit, Theorem B — equivalently Corollary 11.1 — excludes every nontorsion dilation equation, and Theorem 10.2 excludes the mixed equations at every q with $v(q) \neq 0$. Rational coefficient equations reduce to the polynomial-coefficient case by clearing denominators. $\square$

The universal quantifier over q matters. The parameter may be chosen after F is fixed and may itself have infinite support; each individual $v(q)$ still has finitely many coordinates in Γ_∞ , hence is noncofinal. So the theorem excludes a linear relation for *each* such parameter, not merely for an advance list of simple dilations. The series has a simple explicit coefficient rule, and this is a statement about finite linear annihilating equations, not a claim that the rule is uncomputable or lacks a finite description.

11.2 Why the same formula is not entire on all surcomplex numbers

Let $e_* = \omega^\omega$, which dominates every integer multiple of every positive element of Γ_∞ , and enlarge the group to $\Gamma_\infty + \mathbb{Q}e_*$. At the surcomplex point $x = \omega^{e_*} = t^{-e_*}$ the evaluated leading exponents are $e_n - ne_*$, whose successive differences $e_{n+1} - e_n - e_*$ are negative. The defining family is therefore not strongly summable, by Lemma 2.2.

This explicit obstruction prevents the fixed-workspace theorem from being misread as an assertion of a nonpolynomial power series entire on all of $\mathbf{No}[i]$. The example illustrates two independent restrictions: a workspace can be large enough that no single dilation scale is cofinal, yet small enough to admit a countable sequence of coefficient scales escaping all its arguments, and enlarging it further can destroy the very entirety on which the theorem operates. The companion report studies this kind of boundary in general [12]; the calculation here is included only to state the domain of the present example unambiguously, and no new universal scalar-extension theorem is claimed.

11.3 Two classification tables

Let $\Gamma = \mathbb{Q}H \oplus \mathbb{Q}\epsilon$ with H dominating every $n\epsilon$, so that H is an order unit and ϵ is not.

Dilation q	Its valuation	Nonpolynomial entire solution of a nonzero linear q -difference equation?
t^H	H	Yes: Θ_{t^H} , by (8.3).
t^{-H}	$-H$	Yes: Θ_{t^H} , by (8.7).
t^ϵ	ϵ	No: positive, but noncofinal.
$1 + t^H$	0	No: an infinite-order unit.
2	0	No: complex size does not change the Hahn valuation.
-1	0	Outside the theorem: a torsion parameter, with periodic-support exceptions.

Workspace group	Parameter	Solution exists?	Reason
$\mathbb{Q}$	$v(q) \neq 0$	Yes	Every positive nonzero scale is an order unit.
$\mathbb{Q}$	$v(q) = 0$, nontorsion	No	Unit-orbit valuations are eventually periodic.
$\mathbb{Q}e_0 \oplus \mathbb{Q}e_1$, $e_1 \gg e_0$	$v(q) = e_0$	No	A lower scale is not cofinal.
the same group	$v(q) = e_1$	Yes	The top scale is an order unit.
$\bigoplus_{n \geq 0} \mathbb{Q}e_n$	any nontorsion q	No	No scale is an order unit, although F is entire.

In every row, nonpolynomial strongly entire D-finite series are impossible. Torsion parameters are deliberately absent from the second table: their annihilating operator may be an identity on every series. Finite valuation rank alone is therefore not the criterion, and the distinction persists for nonmonomial dilations, because the theorem uses the valuation rather than a requirement that q be a monomial.

Corollary 11.3 (Detecting order units by functional equations). *For a nonzero set-sized ordered abelian group Γ the following are equivalent:*

- (i) Γ has an order unit;
- (ii) some nontorsion $q \in K_\Gamma^\times$ admits a nonpolynomial strongly entire solution of a nonzero polynomial-coefficient linear q -difference equation.

Proof. Theorem B gives (ii) $\Rightarrow$ (i). For the converse take $q = t^\mu$ for an order unit μ ; then $v(q) = \mu$ is an order unit and q is nontorsion. $\square$

This recovers a property of the ordered group from a global functional equation existence question. The workspace (11.1) shows why the existence of nonpolynomial entire functions is by itself strictly weaker than the existence of an order unit.

11.4 Several meanings of transcendence, kept apart

For the explicit F the established conclusions are: transcendence over $K_{\Gamma_\infty}(z)$ by Corollary 5.4; linear differential independence of every finite family of derivatives; and linear independence along every single nontorsion geometric dilation orbit. These are substantial restrictions, and they do *not* imply that no polynomial relation $P(z, F, D_z F, \dots, D_z^R F) = 0$ exists. Nonlinear differential equations lie outside these proofs, and a relation involving $F(q_1 z)$ and $F(q_2 z)$ for multiplicatively independent parameters is not covered merely by applying the single- q theorem twice.

12 Coefficient fields, degenerate value groups, characteristic

The one-variable proofs work over any characteristic-zero coefficient field with trivial valuation: the root-of-unity argument uses only finitely many ordinary polynomial roots and needs no algebraic closedness, and the integer-evaluation lemma needs only residue characteristic zero. All theorems also hold over $\mathbb{R}((t^\Gamma))$, the generic-line proof of Theorem C using uncountability of $\mathbb{R}$ in place of uncountability of $\mathbb{C}$. The results therefore concern surreal as well as surcomplex workspaces. More generally the core versions work over $k((t^\Gamma))$ for any characteristic-zero field k , and, as Section 1.4 records, without divisibility of Γ .

No statement here is a theorem for arbitrary valued fields of positive residue characteristic. In that setting Lemma 4.1 fails and the valuations of near-torsion geometric powers can behave differently, so any comparison with the ultrametric literature must keep the coefficient field and the residue characteristic visible.

We assumed $\Gamma \neq 0$ in order to speak of positive exterior scales. When $\Gamma = 0$ the field is $\mathbb{C}$ with the trivial valuation, strong evaluation at $x = 1$ already permits only finitely many nonzero coefficients because every nonzero term has support $\{0\}$, and entire series are polynomials for that immediate support reason.

13 Quantitative certificates and symbolic representation

13.1 What one can extract from an annihilating equation

The classification has a direct representational consequence. A system whose nonpolynomial special functions are represented only by finite linear differential equations cannot cover the nonpolynomial strong entire functions of a Hahn workspace. Allowing linear dilation equations enlarges the class only when a cofinal dilation scale is available. In the no-order-unit example, an exact coefficient rule such as (11.2) lies outside both pure equation formats.

This does not rule out exact symbolic computation with such functions. It specifies what a faithful representation must keep separate: the coefficient-generation rule, the exponent group, the

strong evaluation domain, and the type of finite equation being asserted. A field-independent label such as “entire theta function” discards relevant information; for Θ_p the proof certificate is not merely $v(p) > 0$, it includes that $v(p)$ is an order unit of the selected group.

For a differential operator the reduction yields a concrete certificate. Compute the finite shift polynomials Q_s of (5.2) and shift the least active index to zero. For each resulting P_j record its least coefficient valuation β_j and the corresponding nonzero complex residue polynomial $\overline{P_j}$. Record an integer N beyond their nonnegative integer roots and beyond the recurrence’s starting index. Record the threshold B of Proposition 5.2, or (5.4) when Γ is divisible. The escape statements then certify that every formal solution strongly evaluable at some $x \neq 0$ with $v(x) \leq -B$ has degree less than N .

For a valuation-zero dilation the same plan uses (7.1): if $\text{res}(q)$ has infinite order use the minimum coefficient valuation, and if it has order h use (4.4)–(4.6) on each residue class. For a nonzero-valuation dilation, finitely many affine comparisons select the eventual winner of each $R_h(q^n)$; project their constants modulo $\Delta_{|v(q)|}$ to obtain the coarsened bound (7.3), or (7.4) when Γ is divisible. When the parameter scale is noncofinal one can instead display $\gamma \gg |v(q)|$ and the explicit bad argument $t^{-(B+\gamma)}$ of Proposition 7.3.

These are mathematical certificate schemas relative to exact coefficient data and order comparisons. They are *not* a terminating algorithm for arbitrary encoded surreals. Deciding whether an arbitrary complex residue is a root of unity, comparing arbitrary surreal values, extracting arbitrary supports, or deciding whether an abstract scale is cofinal can require information not effectively available in a chosen representation, and none of this has been made computable by the proofs.

13.2 A valuation recurrence with zero coefficients allowed

Example 13.1 (Jump lengths, not just jumps). Suppose a sequence that is not eventually zero satisfies

$$t^A a_n + (n-2)t^B a_{n+1} + t^C a_{n+3} = 0 \quad (n \geq 0),$$

with $A, B, C \in \Gamma$. The middle coefficient vanishes at $n = 2$, a finite exceptional index; beyond it the three valuations are A, B, C . The escape chain may move forward by one or by three, and no assumption that all a_n are nonzero is needed. Corollary 3.4 excludes the region $v(x) \leq -\max\{A - B, A - C, 0\}$ under Convention 1.1 alone. If Γ is divisible, Corollary 3.5 gives the sharper excluded region

$$v(x) \leq -\max\left\{A - B, \frac{A - C}{3}\right\}.$$

The maximum must be divided by the actual jump length: replacing every jump by one need not preserve the intended threshold, especially when the valuation gaps have different signs. This is a worked instance of item (1) of Section 1.4.

13.3 What a Lean implementation would have to establish

Beyond the existing support lemma, the most reusable formalization target is Lemma 3.1 together with Theorem 3.3, separated from the rest of the analysis. They require a discrete index sequence, a valuation into an ordered additive group, and a finite-successor selection; the contradiction then reduces to Lemma 2.2, that is, to the impossibility of a nonincreasing sequence of leading exponents in a strongly summable family.

The next targets are the two valuation stabilization results. The integer case (Lemma 4.1) is finite polynomial reduction. The unit-dilation case (Theorem 4.4) is a finite binomial expansion plus a leading-coefficient polynomial on each residue class. Only after these does one need the operator-to-recurrence conversions (Lemmas 5.1 and 7.1), which are elementary. The support monoid of Lemma 2.1 (2) is a separate and independently useful target, needed for Theorem 8.1. The several-variable extension adds countable polynomial avoidance and generic restriction of a finite rational differential system; it need not be part of a first implementation stage.

The ledger `docs/FORMALIZATION.md` now maps part (2) of Lemma 2.1 to the existing positive-support and finite-word theorems in `Surreal/HahnSeries/NeumannWords.lean`. The other parts of that lemma and the main holonomic theorem package remain unformalized. In particular, the new inward-stability lemma and strengthened torsion equivalence are written proofs, not Lean results. The Python checks establish none of these infinite statements.

14 The two finite check suites

Both source manuscripts shipped an independent exact-arithmetic suite, both were run on copies and passed, and neither supersedes the other: they cover overlapping but distinct finite facts, and they are kept side by side in `code/` for that reason. Both use only the Python standard library, integers, and `fractions.Fraction`, with no floating point, no network access, and no third-party packages.

`code/08-finite-recurrences-verify.py` — 3,705 checks, 0 failures. It emphasises the operator-to-recurrence conversions in all three formats and the higher-rank ordered-group arithmetic.

Finite calculation	Checks
Differential operator to coefficient recurrence	648
Dilation operator to coefficient recurrence	336
Mixed operator to coefficient recurrence	276
Unit residues, parity, and finite exceptional cancellation	400
Partial theta identities and finite quadratic estimates	552
Higher-rank lexicographic valuation examples	1,292
Sparse Airy recurrence and finite escape edges	201
Total	3,705

The operator tests compare the grouped recurrence formulas against direct finite polynomial differentiation, dilation and multiplication; the theta tests compare Laurent monomial exponents exactly, including the q^{-1} -to- q conversion of (8.7); the unit tests retain the exceptional index $n = 7$ rather than silently checking only generic values; and the ordered-group checks use lexicographic integer tuples, not a real-valued surrogate norm. This script takes a *required* `--output` path and exits nonzero without one, which is deliberate: it prevents a re-run from overwriting the delivered record.

`code/09-entire-hahn-holonomic-verification.py` — 2,043 checks, 0 failures. It emphasises the boundary behaviour of the exclusion bounds and the exact theta domain.

Finite calculation	Checks
Theta homogeneous equation, positive and negative valuation	92
Theta first-order coefficient recurrence	45
Dilation and differential forward recurrence conversions	72
Unit orbit: parity, exceptional cancellation, residue polynomial	504
Rank-two noncofinal affine minimum	200
Theta excluded upper-scale argument	200
Theta included coarse boundary	170
Theta cofinal-scale growth	120
Finite-coordinate entire witness growth	480
Sparse recurrence exact identity	50
Escape boundary: constant leading exponent, strict exterior descent	50
Formal exponential recurrence	60
Total	2,043

Two of these rows exist only in this suite and check the two boundary phenomena that the report insists are different: “theta included coarse boundary” checks that a zero coarse valuation coexists with positive growth in the original lower-scale coordinate, which is Section 8.5, and “escape boundary” checks the constant-leading-exponent case that Lemma 2.2 rules out, which is what makes the excluded region in Corollary 3.5 closed. Its `--output` argument is optional. Seeds are recorded in the shipped records.

What the checks do not do. Neither suite proves any infinite theorem. No finite prefix can prove strong summability of an infinite family, cofinality of (e_n) , noncofinality of $(n\epsilon)$, the existence of an infinite escape path, the eventual orbit statements, the generic-line theorem, or the absence of every possible annihilating equation. No infinite Hahn sum is evaluated by either program. In particular a finite truncation of the explicit series (11.2) is a polynomial and is D-finite, so no finite computation can witness its nonholonomicity. What the suites can detect is sign, indexing, conversion and finite valuation-comparison mistakes. Everything else rests on the written arguments.

The mathematical dependency chain is short.

Conclusion	Load-bearing ingredients
Differential rigidity	Integer evaluation lemma; finite shift conversion; escape chain and exclusion criterion.
Unit-dilation rigidity	Residue polynomial; finite binomial leading terms; escape chain and exclusion criterion.
Noncofinal nonunit rigidity	Eventual order of finitely many affine values; noncofinality witness; escape chain and exclusion criterion.
Coarsened exclusion	The same affine values, projected to $\Gamma/\Delta_{ v(q) }$; strictness of the projected descent.
Positive dilation existence	Support monoid certificate and the eventual strict increase of h_n ; or, in the order-unit case, quadratic leading valuations and the cofinal-valuation criterion.
Several-variable rigidity	Countable polynomial avoidance; finite-block regrouping; rational differential system on a generic line.

All supports are sets. All operators are finite sums. All integer indices are ordinary nonnegative integers. The coefficient valuation is trivial in characteristic zero. The equations use the external formal variable derivative. None of the proofs asserts convergence in the fine topology of $\mathbf{No}[z]$, substitutes a real-valued norm for an arbitrary-rank valuation, or transports a fixed-workspace evaluation to every surcomplex argument.

15 Originality assessment and questions left open

15.1 Established background and proposed contribution

The D-finite/P-recursive correspondence [1], ordinary properties of Hahn fields, the Hahn–Neumann support lemmas and Higman’s finite-word theorem [6], finite polynomial root bounds, the Conway normal-form identification [9, 10], and partial theta series with their elementary functional identity [5] are established material. The companion report’s cofinality and scalar-extension work, including the group Γ_∞ and the distinction between order units and countable cofinality, is prior project material [12]. The rank-one quadratic-exponent entire example is already in the collection [14]. We claim to originate none of these, and we do not claim to have disproved any published theorem about a different analytic function class.

The proposed research contribution is their combination into the sharp arbitrary-rank dilation classification, Theorem B: in particular the necessity argument for all noncofinal scales, the unit case with root-of-unity residue, the uniform exterior obstruction applying to every solution of a fixed equation, the quantitative exclusion bounds with their two different boundary behaviours, and the exact theta evaluation domain for an arbitrary Hahn parameter. The mixed noncofinal extension (Theorem 10.2), generic-line rigidity (Theorem C), and the explicit series excluded from every nontorsion pure dilation equation (Theorem 11.2) are further consequences proved here.

15.2 A result-by-result attribution ledger

Result or ingredient	Status in this report
Hahn arithmetic, strong support sums, surreal normal forms	Standard background; references and support proofs supplied.
Higman’s finite-word theorem	External standard ingredient; alternatively import the classical Hahn–Neumann support lemma directly.
D-finite / polynomial recurrence conversion	Classical; credited to Stanley and derived explicitly for Hahn coefficients.
Partial theta series and functional identity	Classical; not a novelty claim.
Escape chain and exclusion criterion, with the local-finiteness boundary	Elementary support-theoretic lemma and quantitative formulation developed here.
Eventual affine and periodic orbit valuations	Proved here for arbitrary Hahn coefficients; the finite residue-polynomial treatment covers near-torsion units.
Exact fixed- q order-unit classification	Principal proposed contribution: arbitrary rank, arbitrary Hahn parameter, both signs of $v(q)$.

Result or ingredient	Status in this report
Exact theta evaluation domain	Support-controlled sharpness theorem; related in spirit to the companion report's scalar-extension results, proved independently here.
Uniform exterior obstruction and common degree bound	Proved here; stronger than eventual termination of each solution separately.
Universal single-dilation nonholonomicity on no-order-unit workspaces	Consequence of the classification, with an explicit entire witness.
Mixed differential–dilation rigidity at noncofinal nonzero scales	Proved here; the unit-valuation mixed case is left open.
Several-variable rigidity and multivariate algebraic rigidity	Proved here; the generic line is an existence device.
Named published open conjecture	None claimed solved.
Priority, peer review, formal verification	Priority and independent refereeing are not certified. Only the positive-support word lemma is mapped to Lean; the main theorem package remains unformalized.

The targeted literature checks located the standard D-finite framework, classical complex q -difference growth theory, q -holonomic degree growth in a rational-function setting, and ultrametric q -difference theory, but did not establish whether an identical arbitrary-rank strong-Hahn formulation appears elsewhere. Several broad keyword searches returned irrelevant material and were not treated as evidence of absence; a repository content search for `holonomic` returning no matches is likewise not proof that no sentence anywhere in the repository can be reformulated into one of these results. Accordingly this is a proposed original result with proofs, not a certified claim of bibliographic priority. Independent mathematical review is still needed; machine-checked scope is limited to the support lemma recorded above.

15.3 Further mathematical questions

Question 15.1 (Nonlinear differential equations). Can a nonpolynomial $f \in \mathcal{E}_\Gamma$ satisfy a nonzero nonlinear algebraic differential equation over $K_\Gamma(z)$? Under which hypotheses on Γ can linear rigidity be strengthened to differential algebraic rigidity? Does the explicit series (11.2) avoid all such equations under a natural coefficient restriction?

The escape proof does not answer this. A nonlinear differential relation produces convolution sums in the coefficients rather than a bounded-step linear recurrence, and treating it as though it had the form (3.1) would be an invalid extension.

Question 15.2 (Mixed operators at unit valuation). For a nontorsion q with $v(q) = 0$, must every strongly entire solution of a nonzero mixed equation (10.1) be a polynomial?

The unresolved point in the present method is uniform control of $v(P(n, q^n))$ for a general bivariate P in the unit case. The pure q -difference proof controls $P(q^n)$ and the differential proof controls $P(n)$; neither statement by itself handles their combination. This is a question suggested by the argument, not asserted to be a previously catalogued open problem.

Question 15.3 (Optimal exterior regions and exact solution domains). For a fixed differential or rigid dilation operator, can one compute the largest common strong evaluation domain of its formal solution space from finite valued operator data, rather than only a sufficient exclusion bound? Can the coarse exclusion bound of Theorem 7.4 be supplemented by support hypotheses that give the exact domain?

The thresholds supplied here are uniform and explicit once the valuation certificates are available, but need not be optimal, and distinct solutions may have cancellations or terminating tails that change their individual domains. Proposition 5.3 shows that the uncoarsened bound can be exactly attained, and Theorem 8.1 exhibits one sufficient pattern for an exact answer: a common positive support monoid together with a controlled sequence of leading exponents. A full answer should retain arbitrary Hahn tails rather than use only a numerical radius.

Question 15.4 (Several independent dilations). What is the existence criterion for a nonpolynomial strongly entire solution constrained by equations using several multiplicatively independent dilations? Is an appropriate cofinality condition on their joint valuation data sufficient, or do compatibility relations impose additional obstructions? For a no-order-unit workspace, can one classify the finite linear relations among $F(q_1z), \dots, F(q_sz)$ when the parameters are not powers of a common element?

The single-dilation theorem identifies the exact obstruction when only one scale is repeatedly added. With several dilation directions, distinct monomials can have equal valuation slopes and their leading coefficients can produce nontrivial recurrence phenomena of their own. A meaningful formulation must also quotient out identities caused by repeated parameters and any genuine algebraic relations among them. No multidilation classification is implied by Theorem B.

These questions are directions suggested by the proved statements, not claims that they were already posed in the published literature, and their formulations are deliberately narrower than a general theory of nonlinear surcomplex differential equations.

16 Conclusion

Strong entireness over a surreal Hahn workspace imposes a global scale requirement that a finite linear differential equation cannot meet unless its solution terminates. A dilation equation can meet that requirement, but exactly when its valuation reaches every scale through integer multiples. The decisive distinction is therefore not between Archimedean and non-Archimedean fields, nor between rank one and higher rank; it is between *a cofinal dilation scale* and *a scale trapped below another one*.

The obstruction is finite and robust. A linear recurrence forces a later nonzero coefficient with a controlled valuation, and a large enough argument turns an infinite chain of such coefficients into a nonincreasing sequence of leading exponents. Descending supports and infinite reuse of one exponent are two distinct ways in which the evaluation fails, and keeping them separate is what makes one exclusion boundary closed and the other open. The mechanism works with arbitrary Hahn coefficient tails, arbitrarily high valuation rank, and sparse coefficient sequences, without replacing strong summability by topological convergence and without any hidden completeness assumption.

The countable-rank example makes the separation concrete. The series $\sum_{n \geq 0} \omega^{-\omega^n} z^n$ is evaluable at every point of its full workspace, yet no single nontorsion dilation in that workspace supplies a

finite linear equation for it. That gap between existence and finite linear describability is the main mathematical contribution proposed here.

A Notation

Symbol	Meaning
Γ	A fixed nonzero set-sized ordered abelian group; divisible only where stated.
K_Γ	The complex Hahn field $\mathbb{C}((t^\Gamma))$.
$v(a), \text{lc}(a), \text{res}(a)$	Least support exponent, its coefficient, and the residue when $v(a) = 0$.
$\mathcal{E}_\Gamma, \text{Dom}_{\text{str}}(f)$	Series strongly evaluable at every point of K_Γ ; the exact set where a series is strongly evaluable.
D_z	Formal derivative in z , zero on every element of K_Γ .
σ_q	Argument dilation $f(z) \mapsto f(qz)$.
$M(S)$	The additive monoid generated by a well-ordered $S \subseteq \Gamma_{>0}$.
Δ_β	The smallest convex subgroup containing $\beta > 0$, (2.1).
B, B^+, B_{rec}	Exclusion thresholds in Γ ; B_{rec} divides by the jump length and needs Γ divisible.
$\overline{B}_{\text{rec}}^b, \overline{B}_{\text{rec}}$	Their coarsened analogues in $\Gamma/\Delta_{ v(q) }$.
μ, β	Usually a positive dilation valuation; an order unit only when stated.
$(X)_r$	The ordinary falling factorial polynomial of degree r .
Θ_p	The partial theta series (8.1).
Γ_∞	The increasing-rank direct sum (11.1).

B Proof-dependency and hypothesis audit

The logical dependencies are short. Differential rigidity uses the integer-evaluation lemma, the coefficient conversion, and the escape chain with its exclusion criterion. Dilation necessity uses the geometric-orbit lemma or the unit-orbit theorem, the dilation conversion, and the same criterion. Dilation sufficiency uses only the partial theta identity and the support certificate. The mixed extension uses the bivariate orbit lemma. The several-variable theorem uses countable polynomial avoidance, finite-block regrouping and the one-variable theorem. The infinite-rank application uses a direct strong-entireness proof and the completed classification.

No unproved ring-theoretic conclusion from another report in this collection is used. The real-valued coarsening of [12, Lemma 9.1] is cited and not reproved, and no proof here depends on it. Higman’s theorem is an external standard ingredient; alternatively one may import the classical Hahn–Neumann support lemma directly. The surreal interpretation imports the normal-form field embedding, while every substantive function statement is already proved in the abstract Hahn field.

The potentially dangerous hypotheses, collected for review:

- Supports and the value group are sets; all operators are finite sums; all integer indices are ordinary nonnegative integers.
- $\Gamma \neq 0$, set-sized, ordered abelian. *Divisibility is not assumed* except in Corollary 3.5, the refined threshold (5.4), Theorem 7.4 (b), and Example 13.1, each of which states it.

- Series are ordinary $\mathbb{N}$ -indexed power series in one variable, except in Section 6. Equation coefficients are rational functions of that variable, not arbitrary entire functions; their scalar coefficients may be arbitrary Hahn elements.
- The residue field is characteristic zero and trivially valued. Algebraic closedness is not used.
- Dilation parameters are nonzero and nontorsion; nontorsion is essential, as Section 9 shows by counterexample.
- The derivative is d/dz , not a scalar surreal derivation.
- Entireness is always relative to one fixed workspace, never to $\mathbf{No}[z]$. Strong summability requires both well-ordering and local finiteness.
- The conclusions concern *linear* equations. No nonlinear differential or difference transcendence is claimed, and no unrestricted mixed-dilation independence is claimed.

Two boundary behaviours deserve separate emphasis because they point in opposite directions. The most delicate equality is in Corollary 3.5: weakly decreasing leading exponents already obstruct strong summability, so the excluded region is closed, and Proposition 5.3 shows the bound is attained. The most delicate strictness is in Theorem 7.4: equal *coarse* values need not obstruct summability in Γ , so that excluded region must be open, and Theorem 8.1 with Section 8.5 shows why.

Deliberate exclusions, recorded so that no reader supplies them by analogy: finite-order q is excluded from the main classification and has genuine periodic-support counterexamples; mixed operators at unit valuation are not classified; no nonlinear differential algebraic rigidity is claimed; no complete algorithm for arbitrary surreal comparisons or cofinality decisions is claimed; the finite checks prove no infinite theorem. The main theorem package has no Lean verification or independent refereeing; the existing support-lemma coverage is recorded in the formalization ledger.

C Reproducing the document and the finite checks

The directory contains this standalone L^AT_EX source, its compiled PDF, a README, both source verification programs under `code/`, both recorded outputs and both source build reports under `data/`, and the three source audit documents. No source document from the inspected repository is redistributed. The bibliography is embedded: no BibTeX run, external graphics, shell escape, or repository checkout is required, and a conventional TeX Live or MiKTeX installation with the named packages suffices.

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
python code/08-finite-recurrences-verify.py --output rerun-08.txt
python code/09-entire-hahn-holonomic-verification.py --output rerun-09.txt
```

The `--output` argument is required by the first script, so that a rerun need not overwrite the record distributed with the report, and optional for the second. Both use Python 3.9 or later, exit with a nonzero status if any assertion fails, and use exact integer and rational arithmetic throughout.

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`ent:lem:coarsening`, is the real-valued coarsening cited in Remark 2.6; its Theorem 9.3 is the Bézout statement referred to in Section 1.5.

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